

Multi-timescale neural adaptation underlying long-term musculoskeletal reorganization

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This **important** study investigates how the nervous system adapts to changes in body mechanics using a tendon transfer surgery that imposes a mismatch between muscle contraction and mechanical action. Using electromyography (EMG) to track muscle activity in two macaque monkeys, the authors conclude that there is a two-phase recovery process that reflects different underlying strategies. However, neither monkey's data includes a full set of EMG and kinematic measurements, and the two datasets are not sufficiently aligned with each other from a behavioural point of view; as a result, the evidence supporting the conclusions is **solid** but could be improved.

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Abstract

The central nervous system (CNS) can effectively control body movements despite environmental changes. While much is known about adaptation to external environmental changes, less is known about responses to internal bodily changes. This study investigates how the CNS adapts to long-term alterations in the musculoskeletal system using a tendon transfer model in non-human primates. We surgically relocated finger flexor and extensor muscles to examine how the CNS adapts its strategy for finger movement control by measuring muscle activities during grasping tasks. Two months post-surgery, the monkeys demonstrated significant recovery of grasping function despite the initial disruption. Our findings suggest a two-phase CNS adaptation process: an initial phase enabling function with the transferred muscles, followed by a later phase abolishing this enabled function and restoring a control strategy that, while potentially less conflicted than the maladaptive state, resembled the original pattern, possibly representing a 'good enough' solution. These results highlight a multi-phase CNS adaptation process with distinct time constants in response to sudden bodily changes, offering potential insights into understanding and treating movement disorders.

Significance statement

After major changes to the body's mechanics, the nervous system adapts using strategies on multiple timescales. Our primate tendon transfer study shows that core muscle synergy groupings remain stable, reflecting a default to modular control. However, the activation of these synergies changes dramatically; an initial, rapid 'swap' of their timing proves to be **maladaptive**, impairing motor function. This conflict is only resolved through the gradual development of **slower, compensatory strategies** over several weeks. This process highlights the fundamental tension the CNS faces when its reliance on stable motor modules conflicts with the need for flexible control, offering insights into neural plasticity and staged rehabilitation.

Introduction

The central nervous system (CNS) continuously adapts bodily functions in response to both external and internal challenges. Experimental models based on external perturbations—such as altered gravitational fields or distorted sensory feedback—have illuminated mechanisms of sensorimotor adaptation (Sugita, 1996 [\[1\]](#); Davidson and Wolpert, 2003 [\[2\]](#); Luauté et al., 2009 [\[3\]](#); Fleury et al., 2019 [\[4\]](#)). Because the changes in the external environment can be controlled by the experimental design, these models provide an opportunity to assess how the CNS adapts to them. However, the transient and predictable nature of these changes may not fully capture the demands posed by internal, long-lasting, and unpredictable alterations to the body's internal environment

In contrast, the internal changes such as developmental growth (Power and Schlaggar, 2017 [\[5\]](#)), fatigue (Green, 1997 [\[6\]](#)), postural sway (Zemková, 2022 [\[7\]](#)), and sensory disorientation (Schärli et al., 2024 [\[8\]](#)) impose distinct challenges. Because identifying the source, extent, and time-constant of changes in the internal environment is usually difficult, assessing the corresponding CNS adaptation is also challenging. Particularly, structural alterations to the musculoskeletal system—whether due to injury, disease, or surgery—fundamentally change the body's biomechanics and sensorimotor associations, but the quantification of these changes is usually difficult. Accordingly, the way CNS remaps its motor control strategies corresponding to the changes is not yet well understood (Walker et al., 2004 [\[9\]](#)), although understanding such adaptations is crucial for elucidating the pathophysiology of motor impairments observed in chronic musculoskeletal conditions such as (Hunter and Eckstein, 2009 [\[10\]](#)) and muscular dystrophy (Mercuri and Muntoni, 2013 [\[11\]](#)).

To address this question, in this study, we employed a tendon transfer (TT) surgery model (Sperry, 1940 [\[12\]](#)), which introduces a controlled, sustained change to the musculoskeletal structure. TT surgery is a clinically well-established procedure (Gardenier et al., 2020 [\[13\]](#)) that surgically re-attaches the tendon of a specific muscle to that of a surrounding one. This procedure relocates a specific muscle, so that its contraction generates a new mechanical action and, consequently, novel somatosensory feedback. Because the internal change is controlled by the experimenter, the TT model provides a powerful platform to investigate how the CNS adapts to a new internal state. Another unique feature of TT model is that it places permanent changes of the internal environment while leaving the CNS anatomically intact. Unlike CNS lesion models (Hoffmann et al., 2009 [\[14\]](#))—where the injury itself disrupts neural circuits and thereby complicates the assessment of adaptive capacity of the CNS—TT offers a distinct advantage: it allows investigation of CNS-driven adaptation without the confounding effects of direct neural damage. Tendon transfer has been used in various species, with studies suggesting different adaptive capacities,

ranging from limited adaptation in adult rodents (Sperry, 1940 [↗](#); Sperry, 1942 [↗](#); Slawinska and Kasicki, 2002 [↗](#); Bowlus et al., 2003 [↗](#)) to more substantial adjustments in cats (Yumiya et al., 1979 [↗](#); Loeb, 1999 [↗](#)) and primates, including humans (Lee and Seung, 1999 [↗](#); Wester et al., 2018 [↗](#); Gaetz et al., 2023 [↗](#)). Therefore, adaptability to TT appears enhanced in primates and humans. Altogether, this approach provides a controlled platform to examine how the CNS adapts to musculoskeletal changes in primates.

According to earlier reports, there are two motor control strategies that could be involved when the CNS needs to face the sensorimotor remapping posed by structural alterations to the musculoskeletal system. First, the CNS may employ modular building blocks such as muscle synergies—coordinated activations of muscle groups that reduce the dimensionality of motor control (d’Avella et al., 2003 [↗](#); Bizzi and Cheung, 2013 [↗](#)). Second, skilled behaviors like fine finger movements often require fractionation—the capacity to activate muscles independently. When confronted with structural changes to the musculoskeletal system, does the CNS adapt by modulating existing synergies, or by shifting toward more fractionated control strategies?

This study aimed to identify long-term adaptive mechanisms of the primate CNS following structural changes to the musculoskeletal system. We investigated whether the CNS adapts by modulating existing muscle synergies or by altering the fractionated control of muscles specifically affected by the surgical alteration. We surgically altered the limb structure by performing crossed tendon transfers of finger extensor and flexor muscle tendons. Using a trained finger-grasping task as our behavioral readout, we examined how the CNS recalibrates muscle activity to regain skilled motor function. Our findings provide new insights into the dynamic reorganization of motor control following structural changes of the bodily.

Results

In this study, we developed a novel crossed tendon transfer animal model. This involved surgically swapping the tendons of two antagonistic finger muscles in a macaque monkeys’ forearm (**Fig. 1** [↗](#)). The effectiveness and consistency of the surgery was confirmed by measuring: (i) the distance traveled by muscle fibers and their intramuscular tendons in the transferred muscle (**Fig. 2B, C** [↗](#)); and (ii) the amount of fingertip displacement (**Fig. 2D, E** [↗](#)). Both measurements were induced by percutaneous electrical stimulation over the transferred muscles (**Fig. 2A** [↗](#)). For a detailed description of the results of these measurements see the Methods section. The two monkeys were assigned slightly different tasks, allowing us to examine controlled grasping in Monkey A and a more natural grasp in Monkey B (refer to the Methods section, and **Figs. 3** [↗](#) and **Fig. 4** [↗](#) for further details). The two monkeys ultimately performed related but distinct grasping tasks, a methodological divergence that provided a valuable opportunity to test the generality of the core adaptive mechanisms. Monkey A performed a controlled grasping task requiring a fine precision grip, designed to study adaptation of fine motor control (**Fig. 3** [↗](#)). While the same task was initially planned for both animals, Monkey B performed this task inconsistently, frequently varying its grip strategy, and we could not reinforce this monkey to perform in single strategy consistently. Therefore, to ensure reliable task engagement and in accordance with the ethical principle of Reduction and Refinement (the ‘3Rs’; (Tannenbaum and Bennett, 2015 [↗](#))), the task for Monkey B was modified to a more naturalistic food retrieval grasp that the animal performed consistently (**Fig. 4** [↗](#)).

Once the monkeys had mastered their respective tasks, we recorded control sessions and then performed the crossed TT. Post-surgery, and once the monkeys had fully recovered and were able to perform the task independently (~3–4 weeks), EMG recordings and behavior were resumed.

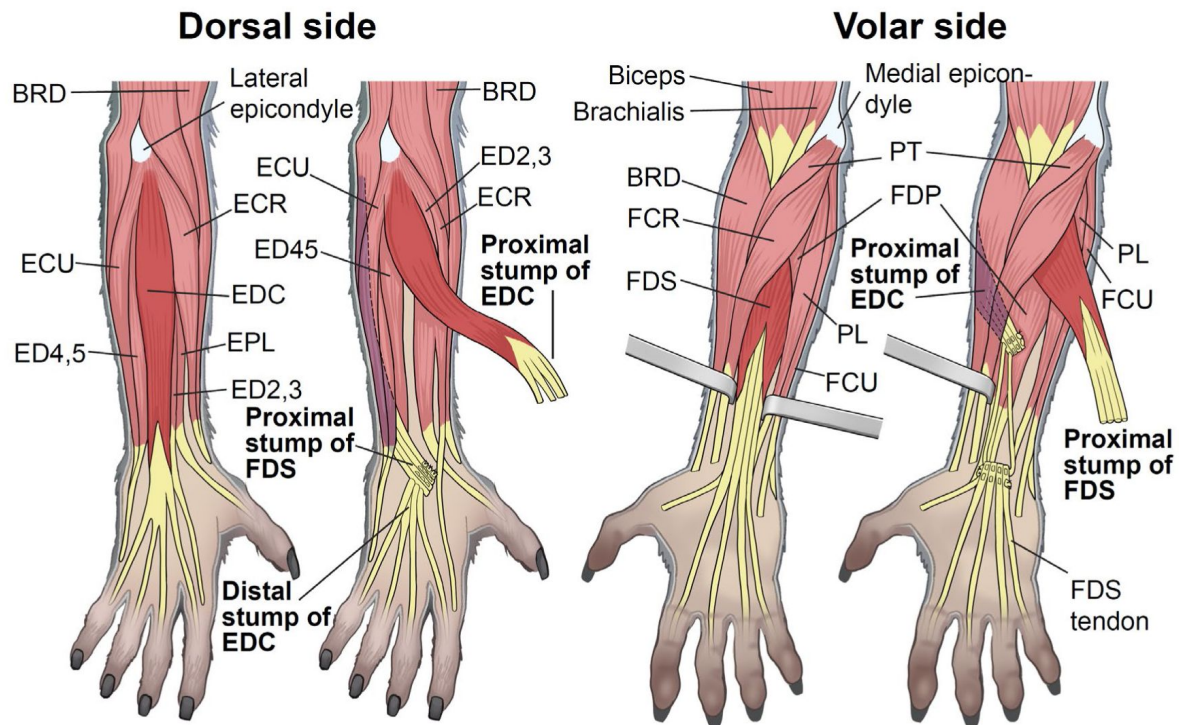


Figure 1

Muscle anatomy of the macaque forearm and the tendon transfer procedure.

A schematic of the primary forearm muscles involved in the study, showing both the dorsal and volar views. The diagram illustrates the surgical cross-tendon transfer of the extensor digitorum communis (EDC) and flexor digitorum superficialis (FDS) tendons. All labeled muscles were implanted with EMG electrodes. Muscle Abbreviations: BRD: brachioradialis, ECR: extensor carpi radialis, ECU: extensor carpi ulnaris, ED2,3: extensor digitorum-2,3, ED4,5: extensor digitorum-4,5, EPL: extensor pollicis longus, FCR: flexor carpi radialis, FCU: flexor carpi ulnaris, FDP: flexor digitorum profundus, PL: palmaris longus, PT: pronator teres, (*The deltoid (DEL) muscle was also implanted in Monkey B but is not shown as it is a shoulder muscle.*)

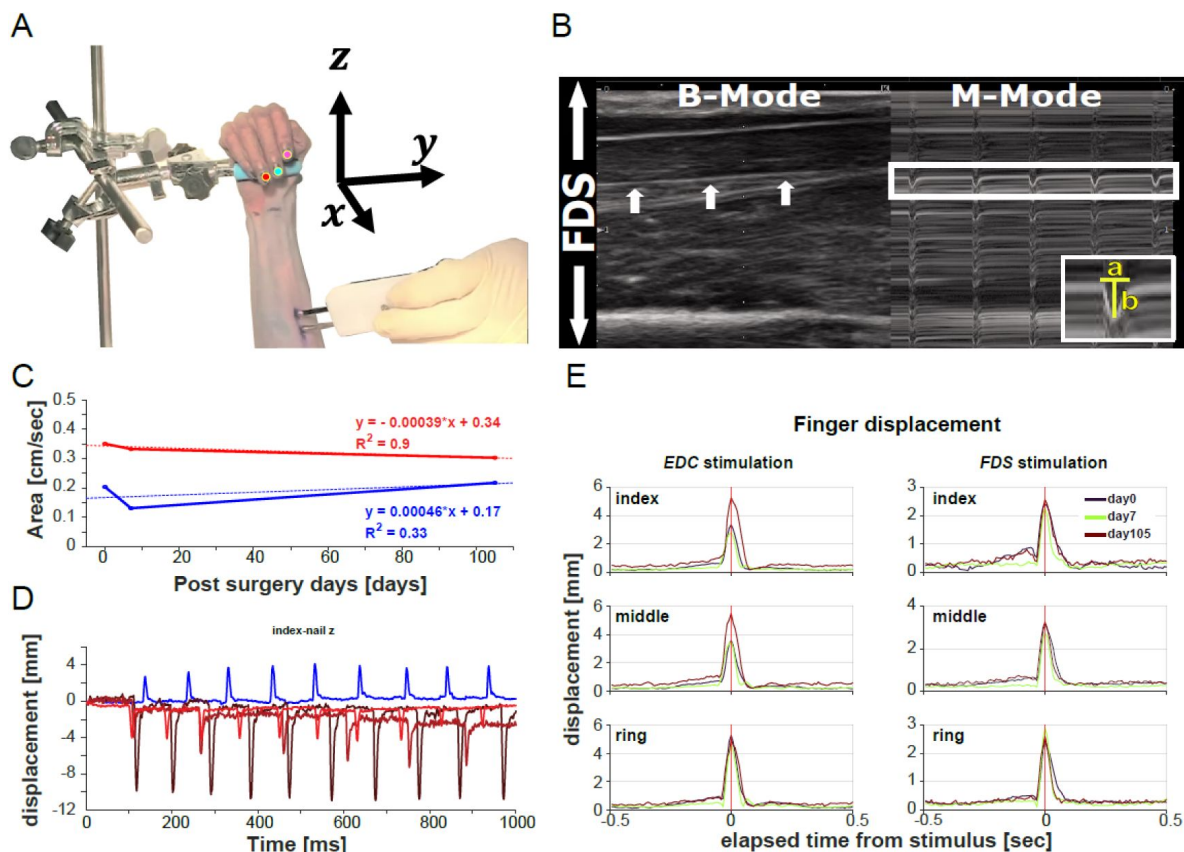


Figure 2

Long term confirmation of tendon surgery effectiveness to alter mechanical properties.

(A) Set-up for the ultrasound measurement and video recordings of the stimulation induced movements of the EDC and FDS tendons. (B) Sonogram of the FDS muscle and its intramuscular tendons. Left side (B-mode, i.e., brightness mode) shows the still image of the monkey's forearm at a given point in time. Right side (M-mode, i.e., motion mode) shows the staggered images of the FDS tendon displacement induced by muscle stimulation (50mA). White arrows demarcate the FDS tendon which was used for the measurement. Grayscale gradations correspond to tissue densities: hyperechoic regions (white) denote denser structures like surface of bones and tendons, while hypoechoic areas (black) signify less dense tissues such as adipose tissue and musculature. Inset demonstrates the area measurement. The area of the displacement waves was measured in the M-mode, representing the strength of muscle contraction. We measured the duration (a , sec) and amplitude (b , cm) of three waves and calculated the average. $\text{Area} = a \cdot b / 2 (\text{cm/sec})$ for days 0, 7 and 105 after tendon transfer. (C) Areas under the wave measured in the M-mode for 3 experimental days (0, 7 and 105 days post-TT) and regression lines in red and blue for FDS and EDC, respectively. $R^2 > 0.5$ for FDS. The data suggested that muscle contractions induced by direct electrical stimulation were nearly constant. (D) Markers placed on the index, middle and ring finger nails (A) were used to measure finger displacement in xyz-dimensions. We calculated the sum of the Euclidean distances of each marker from the origin of the 3D coordinate system as a scalar quantity. Observing the movement along the Z-axis, it became reversed post-surgery indicating a reversal from finger flexion to extension due to tendon transfer (D, blue = pre-TT at surgery day; dark brown = post-TT at surgery day; light brown = 1wk post-TT; red = 3wks post-TT). The scalar quantity of the fingers during muscle stimulation did not change much at day 0, 7 and 105 days (E), suggesting that there was no tendon rupture or slackening of the tendons postoperatively (EDC stimulation, left; FDS stimulation, right).

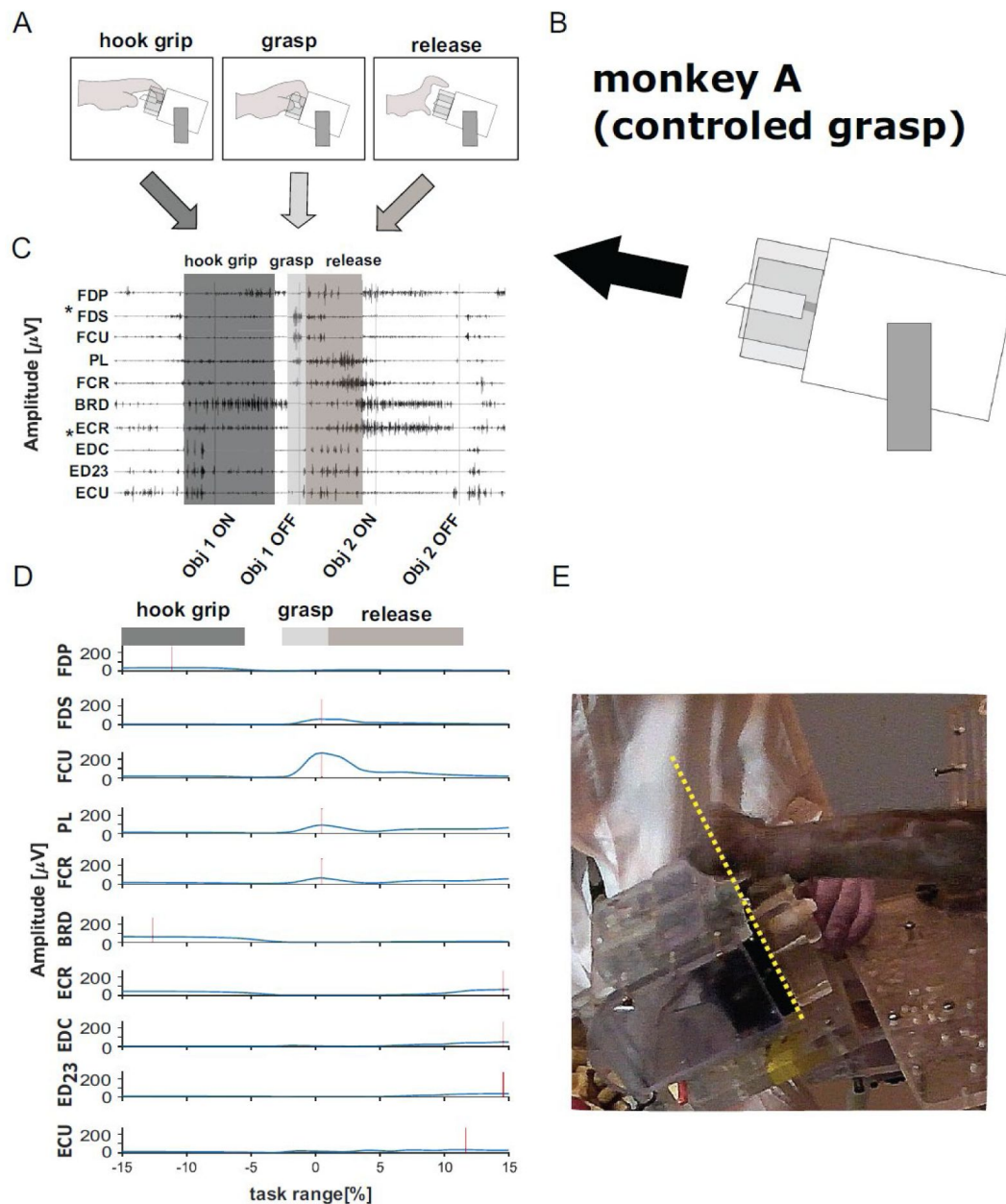


Figure 3

Experimental set-up, task sequence and typical EMG (monkey A).

(A) Schematic of the behavioral sequence (hook → grasp → release). (B) Schematic of the task object using a rod requiring monkey A to perform a controlled grasp. (C) Typical EMG traces of a control session (high-pass filtered) for all recorded muscles. Gray boxes represent the task sequence. Obj 1 ON: start of the hold period of object 1. Obj 1 OFF: end of the hold period of object 1, i.e., object release. Obj 2 ON: start of the hold period of object 2. Obj 2 OFF: end of the hold period of object 2, i.e., object release. Tendons of the muscles marked with * were cross-transferred. (D) Rectified and smoothed EMG for all recorded muscles (average for one recording session; amplitude [μV] over task sequence [%]). Horizontal bars illustrate the corresponding behavioral periods, red vertical lines indicate peak amplitude for each muscle). (E) The time the monkey spent on the left side of the yellow dotted line while moving from object 1 to object 2 was measured and used to quantify the mal-adaptive behavior.

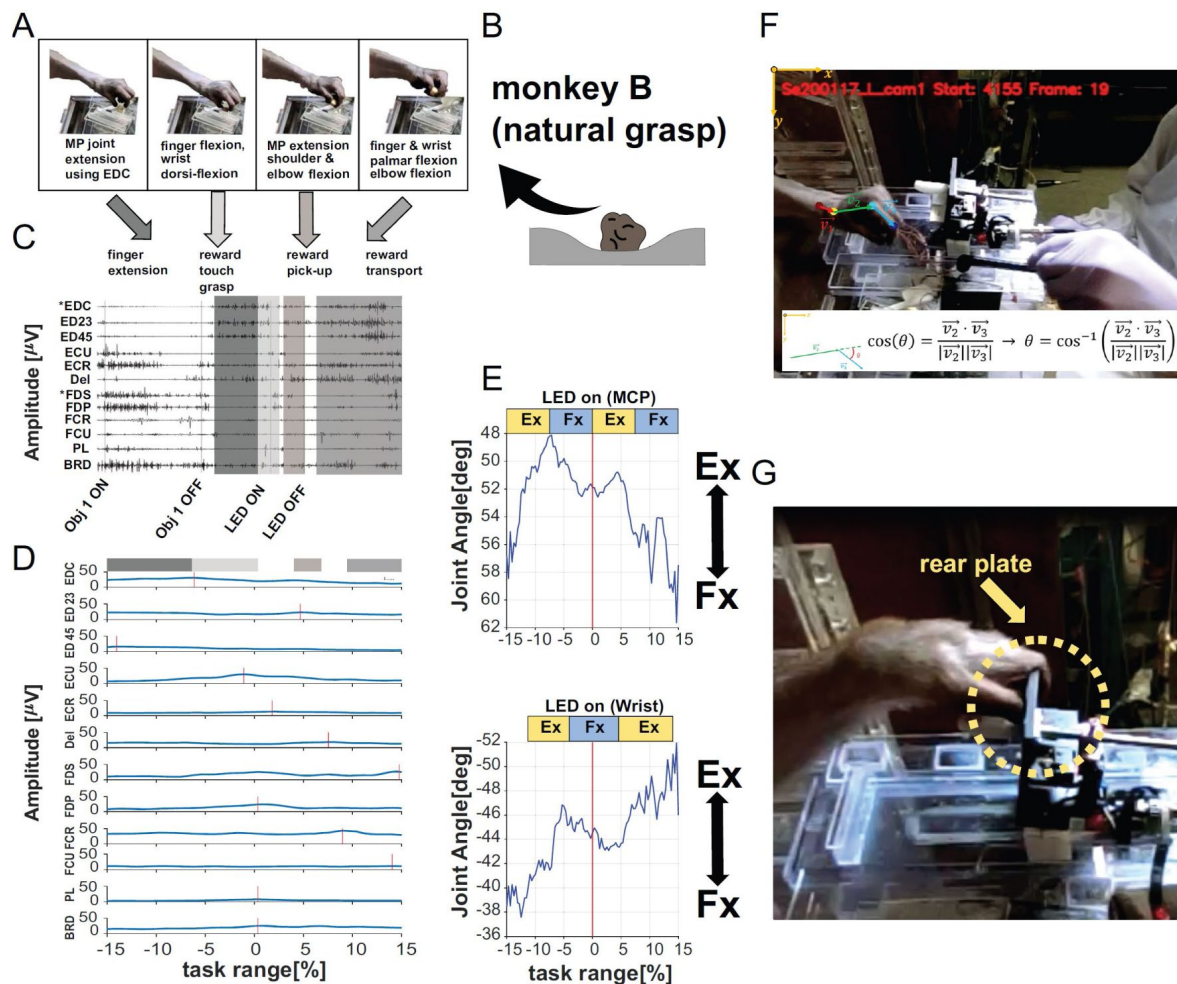


Figure 4

Experimental set-up, task sequence and typical EMG (monkey B).

(A) Schematic of the task sequence (picking up food). (B) Schematic of the task requiring monkey B to pick up food from a groove allowing for a more natural grasp. (C) Typical EMG traces of a control session (high-pass filtered) for all recorded muscles. Gray boxes represent the task. Obj 1 ON: start of the hold period of object 1. Obj 1 OFF: end of object 1's hold period, i.e. object release. LED ON: approximate start of the food touch. LED OFF: approximate time of food retrieval. Tendons of the muscles marked with * were cross-transferred. (D) rectified and smoothed EMG for all recorded muscles (average for one recording session; amplitude [μ V] over task sequence [%]). Horizontal bars illustrate the corresponding behavioral periods, red vertical lines indicate peak amplitude for each muscle. (E) Development of the metacarpophalangeal joint (MCP) and wrist joint angle [deg] over task range [%]. Decreasing and increasing angles indicate extension or flexion of the MCP or wrist angle, respectively. Data are aligned on LED onset (food retrieval) and indicate consistency between behavioral observation (A), EMG (C-D) and kinematics (E). (F) Illustration of joint angle measurement using a still picture of the monkey while performing the task. (G) Example for mal-adaptive behavior in monkey B. The time the monkey spent in contact with or behind the object plate was measured and used to quantify the mal-adaptive behavior.

Functional Recovery Follows a Period of Maladaptive Behavior

Despite the now reversed roles of the transferred muscles, the monkeys were able to recover their grasping performance within two months, assessed by both the return of key behavioral metrics to pre-surgical levels (Fig. 5A) and their consistent ability to perform the task independently (Supplementary Videos S3, S4, S7). However, this recovery was not immediate. The initial post-surgical period was instead defined by a phase of significant *maladaptation*, primarily characterized by aberrant reaching. These aberrant movements included inefficient, ‘explorative’ trajectories in Monkey A and target overshoots in Monkey B (Supplementary Videos S2, S3, S6). This impairment was quantified across two key behavioral metrics: grip formation times (Fig. 5A) and aberrant reaching times (Fig. 5B). First, task-related grip formation times were significantly longer immediately post-TT (Monkey A: 197.7 ± 92.2 vs. 660.6 ± 221 ms, $p = 0.014$; Monkey B: 169.7 ± 14 vs. 316.3 ± 31.9 ms, $p = 0.016$); it took approximately 10 days for this metric to return to pre-TT levels and 20 to 32 days, respectively, to stabilize. Second, the duration of the aberrant reaching component itself was substantially elevated post-surgery. Monkey A exhibited these movements from day 29 (1978 ± 715 ms) to 73 (93.4 ± 3.7 ms), while Monkey B showed them from day 20 (103.4 ms) to 62 (107.6 ms), with the duration peaking at days 38-46 (167 ± 1.4 ms). Ultimately, overcoming these maladaptive behaviors restored movement efficiency and precision, leading to the stabilization of grasping performance by the end of the experimental period.

A two-phase adaptation is observed in the EMG activity of individual muscles

Initially, we investigated whether EMG activity exhibited any changes post-TT. Given that we interchanged the tendons of an antagonistic muscle pair, functionally rendering the FDS an extensor and the EDC a flexor (as confirmed mechanically, Fig. 2), a biomechanically sound adaptation would be to use the former finger extensor (EDC) for finger flexion, and the former finger flexor (FDS) for finger extension during a grasping task. We therefore analyzed EMG activity from both transferred and non-transferred muscles to determine if these expected functional changes occurred. Figure 6 shows a comparison of EMG activity profiles for the two transferred muscles, EDC and FDS, in Monkey A (Fig. 6A-D). Prior to surgery, the control showed distinct and contrasting EMG profiles in both muscles. For instance, peak activity in EDC (▼) occurred long after completion of the grasping action (indicated by vertical dashed lines at 0% task range) at 15% task time (Fig. 6A), coinciding with the animal pre-shaping its hand to prepare for the subsequent grasp. Whereas peak activity in FDS (▽) was observed immediately after completion of the grasping action (0.49%; Fig. 6C). The question we posed was whether the post-surgery activity of EDC resembled its original profile (without adaptation to surgery) or the activity of FDS (the anticipated profile following surgical adaptation), and vice versa. Our findings suggested that EMG activity patterns largely shifted in a manner consistent with the new mechanical function imposed by the transfer. Early after TT, peak activity of EDC occurred at -0.58% task time (Fig. 6B; days 29 [red line] and 64 [orange]), which aligns with peak activity of FDS prior to TT (Fig. 6C; black line). Also, for FDS, peak activity occurred at 11% task time (Fig. 6D; ▼), which aligned with peak activity of EDC (Fig. 6A).

This was corroborated by the results in Monkey B (Fig. 6, panels K-M). Although the EMG patterns of both muscles varied significantly as the two monkeys performed different types of grasping actions, we found that the EMG of EDC peaked at -6.12% before (Fig. 6K; ▼) and 0.35% and 1.05% after TT (Fig. 6L; ▽, days 22 and 36, respectively), closely matching the timing for FDS control data (Fig. 6M). The corresponding analysis for FDS could not be performed in Monkey B, since the EMG signal of FDS was lost early after TT (see Methods for details). We were able to extend our recordings to 122- and 64-days post-TT for monkeys A and B, respectively. This allowed us to examine whether the adaptations observed in the early period were consistent throughout the experimental period.

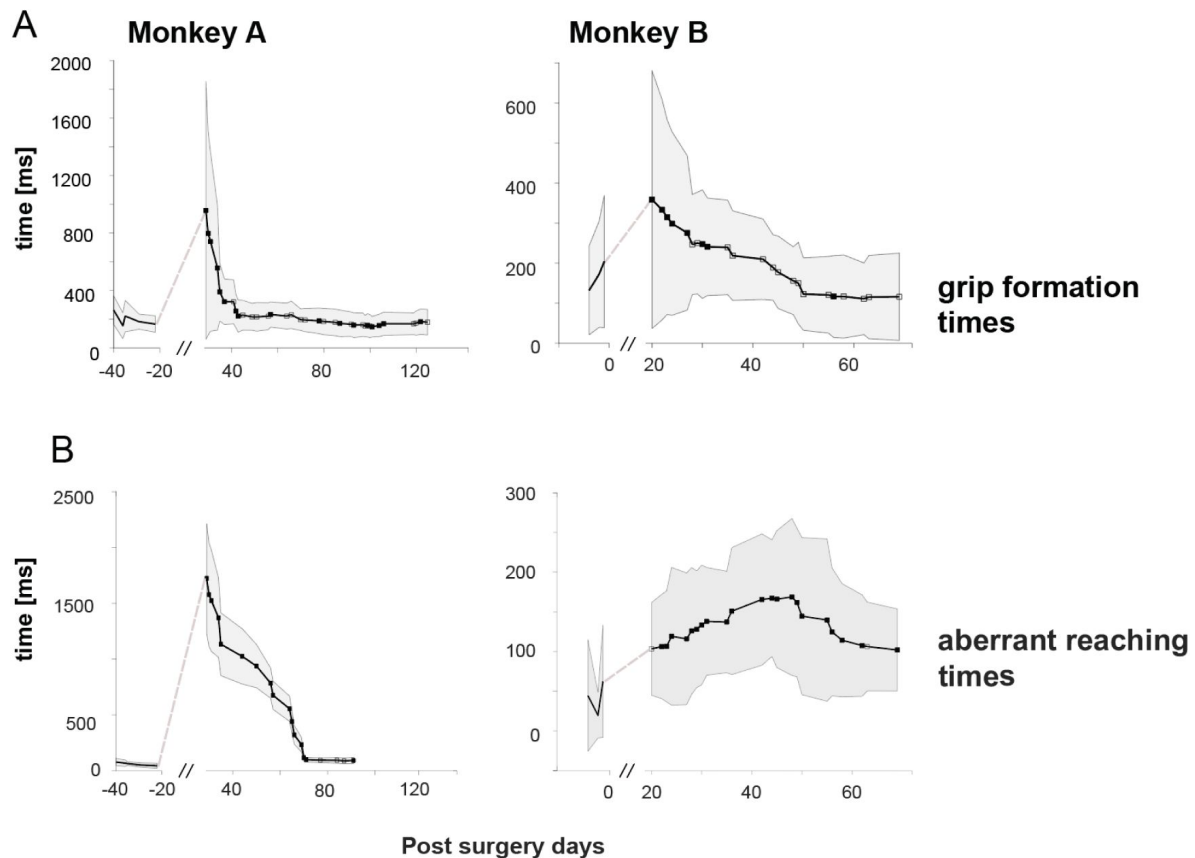


Figure 5

Behavioral recovery is preceded by maladaptive motor control in both monkeys.

(A) Grip formation times. Plotted is the average time (mean $\pm$ SD, $n=20$ trials) from initial touch to action onset (pull or lift) for Monkey A and Monkey B. Grip formation times were significantly elevated immediately post-surgery before returning to baseline. (B) Quantification of aberrant reaching. Plotted is the average time (mean $\pm$ SD, $n=10$ trials) spent executing aberrant movements. For Monkey A, this was the duration spent moving behind the target (see Fig. 3E); for Monkey B, it was the duration in contact with the object's rear plate (see Fig. 4G). These behaviors were prominent in the early post-surgical phase and diminished over time. Filled squares indicate a significant difference from pre-TT baseline ($p<0.05$, two-sample t-test). All data are plotted over days relative to the tendon transfer (TT) surgery (day 0).

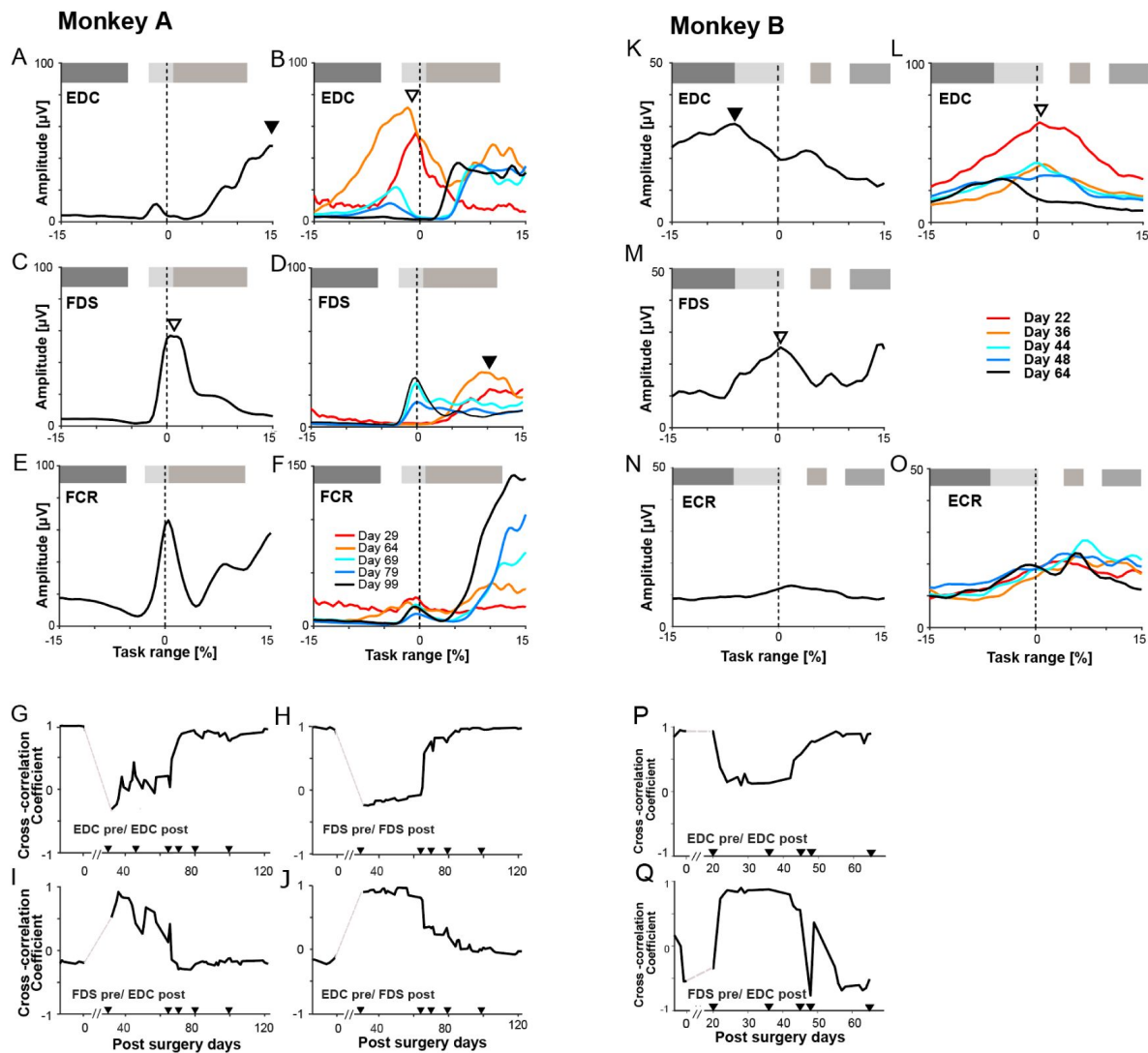


Figure 6

A Two-Phase Adaptation of Muscle Activity is Observed in Both Monkeys Following Tendon Transfer.

The figure compares EMG activity profiles and cross-correlation analyses for Monkey A (left) and Monkey B (right), demonstrating a consistent pattern of motor adaptation consisting of an initial functional swap followed by a later reversion. A representative non-transferred muscle is included for each monkey. Monkey A (A-J): Profiles are aligned on object release. Monkey B (K-Q): Profiles are aligned on food touch. Filled triangles (▼) indicate peak activity occurring during finger extension, while open triangles (▽) indicate peak activity occurring during finger flexion. Small triangles on correlation plots denote landmark days.

These additional days (**Fig. 6B, D**; 69 [cyan], 79 [blue], and 99 [black] post-TT; see Fig. S1 for recordings from all days) showed that the EMG activity profiles for both muscles unexpectedly returned to their pre-surgery state. For example, as already described, the EMG signal of EDC at days 29 and 64 shared the characteristics of FDS, but by the following day (only 5 days later, day 69 [light blue] post-TT) already exhibited pre-TT characteristics of EDC, continuing as such until the end of the recording period (day 79 [blue] and 99 [black]). This was confirmed for the EDC muscle of Monkey B (**Fig. 6K, L**; Fig. S1U-AO for recordings from all days).

Figure 6G illustrates cross-correlation coefficients between the EDC EMG profile post-TT (EDC-post) and the original EDC control data (EDC-pre) in Monkey A. **Figure 6I** shows coefficients for EDC-post and FDS-pre correlations in the same monkey. The coefficient corresponding to the original profile decreased to between -0.3 and 0.2 over approximately 65 days (**Fig. 6G**). In contrast, the coefficient corresponding to the expected EMG profile exhibited an inverse pattern, with low coefficients pre-TT and a high coefficient (of 0.9) post-TT (**Fig. 6I**). Cross-correlation analysis for FDS (**Fig. 6H, J**) revealed a similar pattern. Cross-correlation coefficients for FDS-pre-vs. FDS-post-analysis declined from 1 to negative values, only to rebound to their original values within a month (day 79; **Fig. 6H**). This outcome aligns with our previous observations (**Fig. 6B, D**), and were corroborated by the results from Monkey B (**Fig. 6K-Q**). Here, cross-correlation coefficients dropped from $+0.95$ to $+0.1$ for EDC-pre vs. EDC-post comparison, and increased from -0.5 to 0.9 for the FDS-pre-vs. EDC-post comparison (**Fig. 6P, Q**). After 42 days post-TT, cross-correlation coefficients started to gradually increase or decrease to $+0.9$ and -0.7 , respectively. This indicates that the anticipated initial changes in EMG activity profiles are not permanent. Instead, they revert to their original patterns two months post-TT. Interestingly, changes in EMG pattern were not isolated to the transferred muscles; non-transferred muscles exhibited a variety of complex adaptations (**Figs. 6, S1-S3**). Many agonists adapted in concert with their transferred synergist, following the same two-phase “swap-and-revert” pattern seen in **Figure 6**. In Monkey A, these included extensors ED23 and ECU, while in Monkey B, they included ED23, ED45, and ECU (see Fig. S1 for details). In contrast, other muscles showed patterns that were incompatible with a simple swap. For example, the non-transferred flexor carpi radialis (FCR) in Monkey A showed a distinct adaptive profile characterized by a drastic initial decrease in one activity peak and a consistent increase in a later peak (**Fig. 6E, F**). A similar incompatible pattern was seen in palmaris longus (PL) (Fig. S1H, I, R, S). Finally, some muscles, like the extensor carpi radialis (ECR) in Monkey B, remained relatively stable post-surgery (**Fig. 6N, O**).

Adaptation occurs through modulating the activation of stable muscle synergies

Alternation of EMG activity in non-transferred muscle suggests one possibility: TT might alter the control strategy of coordinated muscle activity for hand movement by modifying the transferred muscles and their agonists as a cohesive unit (i.e., muscle synergy). To test this, we conducted a similar analysis on temporal activation coefficients of muscle synergies. We computed synergy weights (**Fig. 7A, B** and **Fig. 8C, G** for Monkey A; and **Fig. 7E, F** and **8 D, H** for Monkey B) and temporal activation coefficients (**Fig. 7 C, D** and **Fig. 8 E, I** for Monkey A; and **Fig. 7 G, H** and **Fig. 8 F, J** for Monkey B) from EMGs using non-negative matrix factorization (NMF) (Lee and Seung, 1999). We extracted four muscle synergies that accounted for more than 80% (Cheung et al., 2012) of the ‘variance accounted for’ (VAF) (**Figs. 8A, B, 85 %** for both monkeys; the results of the original and shuffled data sets are shown). Among these synergies, one was a finger flexor synergy and another was a finger extensor synergy, identified before the crossed tendon transfer in each monkey (**Fig. 7A, B, E, F** and Fig. S4). These were referred to as synergy A and synergy B, respectively.

We examined if the composition and activation coefficient of each synergy changed, corresponding to the changes observed in the transferred and non-transferred muscles. First, we analyzed changes in the composition of these synergies. The cosine distance for the same synergy

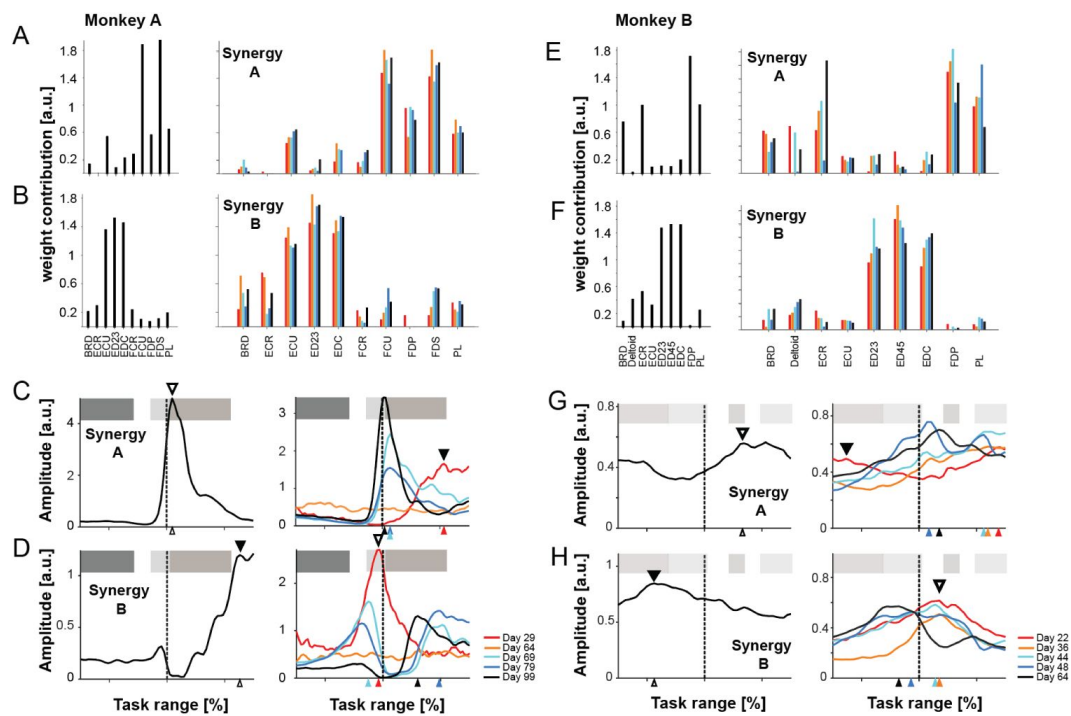


Figure 7

Muscle Synergy Compositions Remain Stable While Their Activation Timings Show a Two-Phase Adaptation.

The figure presents the spatial muscle weights and temporal activation coefficients for the primary finger flexor (Synergy A) and extensor (Synergy B) for Monkey A (A-D) and Monkey B (E-H). (A, B, E, F) Spatial Synergy Weights: The muscle weight contributions for each synergy are shown for the pre-surgery control period and for five post-surgery landmark days, demonstrating that synergy structure was preserved. (C, D, G, H) Temporal Activation Coefficients: The activation profiles show the same two-phase adaptation seen in individual muscles: an initial swap followed by a later reversion. Symbols and alignment are as described in Fig. 6.

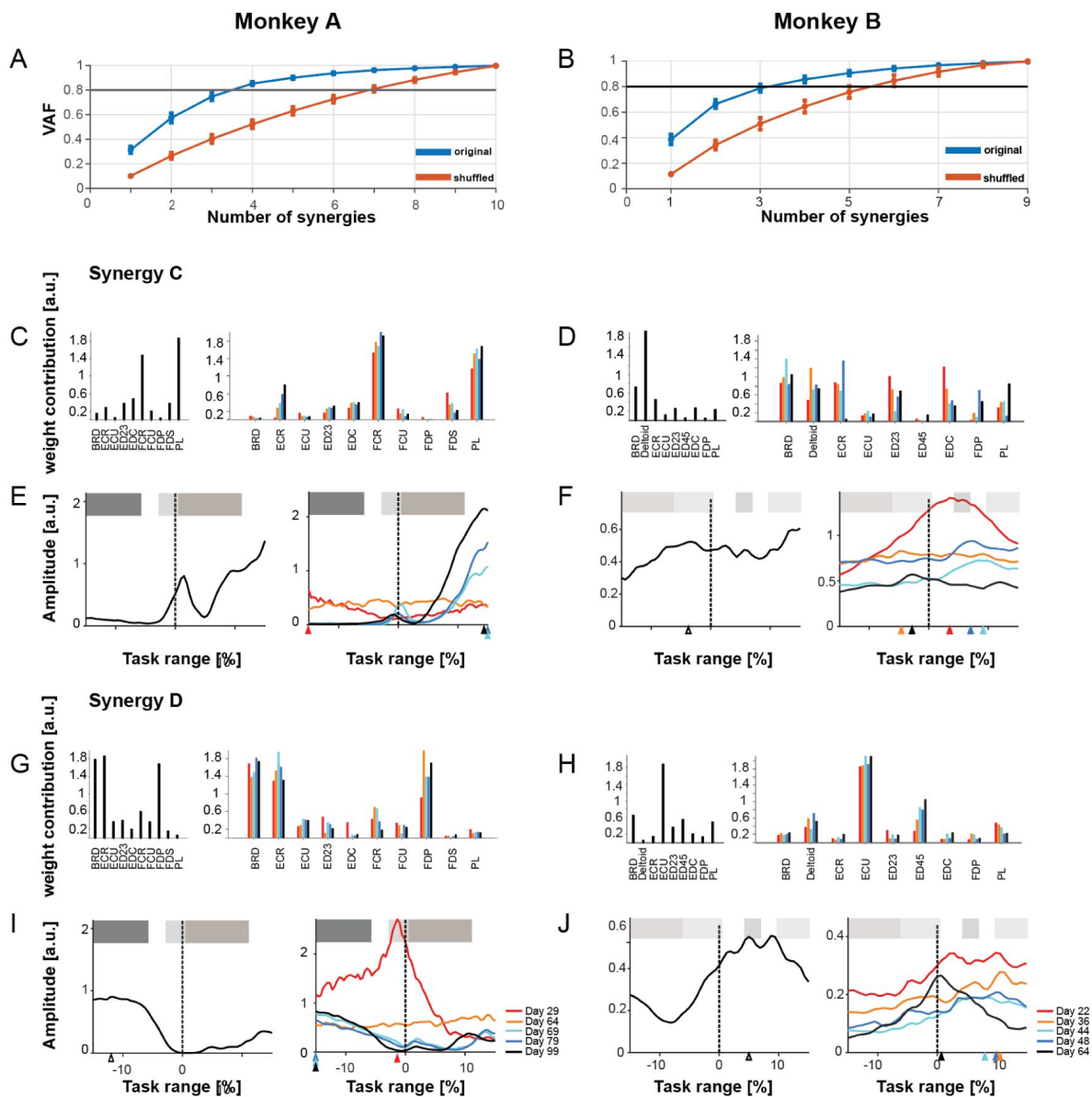


Figure 8

Analysis of Secondary Muscle Synergies and Variance Accounted For (VAF).

This figure provides supporting data for the synergy analysis, including the VAF plots justifying the use of four synergies, and the detailed analysis of the secondary Synergies C and D for both Monkey A and Monkey B. (A-B) Variance Accounted For (VAF). The plots show the cumulative variance in the EMG data explained by an increasing number of synergies for Monkey A (left) and Monkey B (right). In both original datasets (blue lines), four synergies were sufficient to account for more than 80% of the variance (demarcated by the black horizontal line), justifying the dimensionality of the synergy model. VAF for corresponding shuffled data is shown for comparison (red lines). (C-F) Analysis of Synergy C (Wrist Flexor Synergy). This panel group shows the spatial weights (C, D) and temporal activation coefficients (E, F) for Synergy C in Monkey A and Monkey B, respectively. Spatial Weights (C, D): Pre-surgery control profiles (left sub-panels) are compared to post-surgery landmark days (right sub-panels), demonstrating that the muscle contributions to this synergy remained largely stable. Temporal Activation (E, F): In Monkey A, the temporal profile (C) shows a notable increase in activation during the late adaptation phase, consistent with its recruitment for a compensatory strategy. The adaptive pattern for Monkey B (E) was less distinct. (G-J) Analysis of Synergy D. This panel group shows the spatial weights (G, H) and temporal activation coefficients (I, J) for Synergy D in Monkey A and Monkey B. This synergy, which contributes to wrist extension, exhibited relatively minor and inconsistent changes following the tendon transfer in both animals. For all temporal plots, profiles are aligned on object release (Monkey A) or food touch (Monkey B) at 0% task range. Post-surgery plots show data from selected landmark days as indicated by the color legend.

pairs (e.g., synergy A of pre-TT and synergy A of post-TT in all recording days (Fig. S4A, E; monkey A and B, respectively) was low (mean [$\pm$ SD]: 0.03 [$\pm$ 0.03] in Monkey A and 0.09 [$\pm$ 0.11] in Monkey B) and different pairs (e.g., synergy A of pre-TT and synergy B of post-TT) was high (mean [$\pm$ SD]: 0.67 [$\pm$ 0.08] in Monkey A and 0.61 [$\pm$ 0.14] in Monkey B). There was a one-to-one correspondence between synergy pairs ($p < 0.0001$, two-way analysis of variance [ANOVA] due to synergy factor; $p < 0.01$, Bonferroni post-hoc test between synergy of the same pair and different pair). This was demonstrated in both monkeys (S4A-D and E-H for monkey A and B, respectively). These results indicate that the composition of these four synergies and the contribution of each muscle do not change drastically before and after TT surgery.

Second, we analyzed changes in the activation coefficients of these synergies. Prior to TT, the activation coefficients for the flexor synergy (synergy A) reached a peak at +0.97% (Fig. 7C). Post-TT, these coefficients shifted towards later times shortly after surgery (+10.7%), but eventually returned to their original timing. Meanwhile, the extensor synergy (synergy B) peaked at 12.62% pre-TT and moved to -0.97% shortly after surgery. Like the flexor synergy, the extensor synergy also reverted to its original timing in subsequent sessions (Fig. 7D). These characteristics corresponded well to the ones found in the transferred muscles (Fig. 6), supporting our hypothesis that the transferred muscle and its agonist changes as a cohesive unit.

Figure 8E shows the temporal changes for synergy C, a wrist flexor synergy (Fig. 8C and S4C). While two peaks were observed prior to surgery (+1.46% and +15%), the early post-surgery sessions did not exhibit any significant peaks. However, later sessions displayed a single peak at +14.6% task time, with an increasing magnitude as a function of recovery days. These characteristic changes corresponded well to a pattern that was incompatible to the transferred muscles found in FCR and PL, as described above. Indeed, both muscles showed highest synergy weights for this synergy (Fig. 7C and S4C). Synergy D was a wrist extensor synergy (Fig. 8G and S4D), and underwent a single notable change on day 29, but quickly reverted to its original pattern (Fig. 8I). These patterns of changes in synergy 3 and 4, both contributed minimally to the EMG of transferred muscles. This suggests that some non-transferred muscles change their EMG activity following the mode, which is independent to the one observed in the transferred muscle and may reinforce the functional adaptation made by the transferred muscle and its synergists.

Several of these results were also confirmed in Monkey B. Activation coefficients of extensor synergy B peaked at -8.75% before TT, increasing to +3.5% after TT (Fig. 7G). This peak activity persisted until the last day (day 69 post-TT), when it shifted back to an earlier time than -10% of the task range. This pattern corresponds well to that found in the transferred muscles (EDC; Fig. 6K, L).

Again, the characteristics of these changes in activation coefficients of muscle synergies were also quantitatively confirmed by cross-correlation analysis of all four muscle synergies in both monkeys. The coefficients were plotted over the course of experimental days in relation to tendon surgery (Fig. 9). In this analysis, the activation coefficient of each synergy was cross-correlated with either the one from synergy A (finger flexor synergy; Fig. 9C, D, G, H) or synergy B (extensor synergy; Fig. 9A, B, E, F) before TT. As previously demonstrated for the FDS and EDC muscles, the temporal activation coefficients of muscle synergies A and B (flexor and extensor) displayed a distinct pattern. After cross-correlation with their own control data, both the flexor and extensor synergies showed coefficients of 1 prior to surgery. These became negative for 66 days post-surgery before reverting close to their original values (Fig. 9B, C). However, after cross-correlation with control data of the antagonistic muscle synergy (i.e., extensor for flexor synergy and vice versa, Fig. 9B, D), the pattern was reversed. Coefficients started with negative values prior to TT and shifted to positive coefficients near 1 shortly after surgery. Around 66 days later, the correlation coefficients returned to negative values. In short, these results mirror our findings between the transferred muscles (Fig. 6G, H). For synergies C and D, the changes in cross-correlation coefficients over time were markedly different. For synergy C, coefficients begin

to steadily increase when cross-correlated with control data of the extensor synergy (synergy B, **Fig. 10E**), and decreased with the flexor synergy (synergy A, **Fig. 10G**). This suggests an increasing contribution to finger extension. In contrast, synergy D does not exhibit any specific pattern following TT.

In Monkey B, this cross-correlation analysis revealed a more varied, or differential, pattern of adaptation across the four synergies (**Figs. 9** and **10**). The primary extensor, Synergy B, mirrored the results in Monkey A, showing a clear swap-and-revert pattern (**Fig. 9F, H**). In contrast, Synergy A did not show a clear reversal; its correlation coefficients gradually converged towards zero, likely due to the absence of the FDS EMG signal in the analysis (**Fig. 9E, G**). Of the secondary synergies, Synergy C showed no discernible trend (**Fig. 10E, G**), while Synergy D exhibited a complex pattern of an immediate change, a quick reversion, and a second gradual change late in the adaptation period (**Fig. 10F, H**).

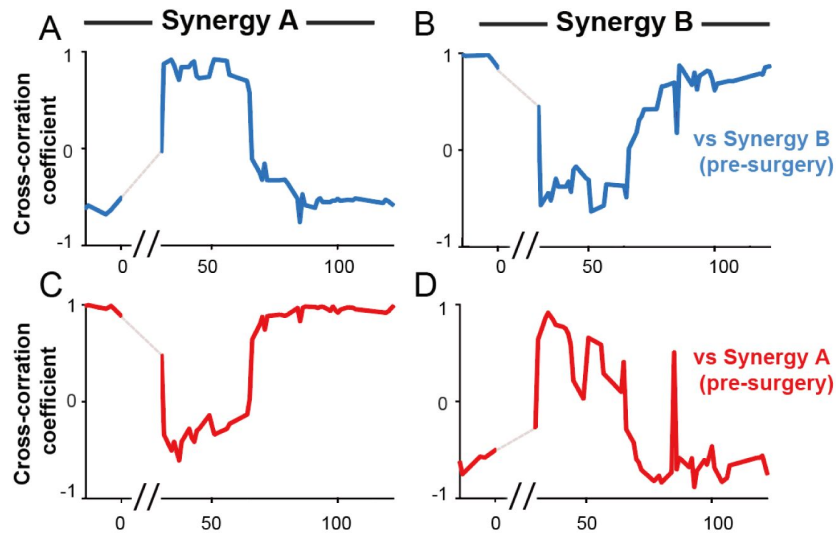
In summary, both monkeys exhibited a distinct two-phase adaptation following TT. We define the ‘early phase’ as the period from initial post-surgical recovery up to the reversal of the swapped activation patterns (approximately days 20/29 to ~65 post-TT, see **Fig. 9**), characterized by the transferred muscles/synergies adopting antagonistic temporal profiles. The ‘late phase’ encompasses the period following this switch-back (after ~day 66 post-TT), where original activation timings were largely restored.

The Early Adaptation Phase is Characterized by a Maladaptive Neural and Behavioral Profile

The early adaptation phase was defined by a distinct neural strategy that correlated with significant behavioral deficits. First, the period dominated by aberrant reaching movements and prolonged grip times (**Fig. 5B**) was precisely when the swapped activation of Synergy A and B was most prominent (**Fig. 9**). This temporal link provides strong evidence that this initial ‘swap’ strategy was, in fact, maladaptive, as the flawed neural control directly underpinned the impairments in movement efficiency and precision.

Second, we found that the net activity of muscles representing certain muscle synergies (aggregated average EMG; aaEMG) showed distinct, synergy-specific changes over time (**Fig. 11**). A key difference was observed in the finger extensor synergy (Synergy B). In Monkey A, this synergy demonstrated what we term a neuromuscular ‘arms race’: a steady and significant increase in aaEMG post-TT until day 64 ($p < 0.0001$) as the CNS appeared to escalate activation to overcome the internal conflict between the transferred EDC (now a flexor) and its non-transferred synergists (which remained extensors), before abruptly reverting to its pre-TT level (**Fig. 11B, F**). In contrast, while Monkey B also showed a significant initial surge in Synergy B activity (day 22: $p < 0.0001$), this was followed by a gradual return to baseline and below (day 64: $p < 0.001$), lacking a sustained ‘arms race’ (**Fig. 11J, N**). This discrepancy is likely attributable to the different force requirements of the tasks. The ‘arms race’ in Monkey A may have been driven by the need to generate high grip force against the synergy’s internal conflict, whereas Monkey B’s delicate food-retrieval task did not require such high forces, thus lacking the same drive for escalating, energetically costly activation. For the finger flexor synergy (Synergy A), Monkey A showed a general decrease in activity ($p = 0.0004$ at day 79; **Fig. 11A, E**), while Monkey B showed a consistent and significant increase throughout the experiment (day 22: $p = 0.0004$; day 64: $p = 0.0008$). This was expected, as Monkey B’s Synergy A relied on FDP and PL to compensate for the loss of FDS function (**Fig. 11I, M**). Synergy C exhibited a general increase in activity during the post-TT phase in both monkeys (**Fig. 11G, O**), while Synergy D showed no significant differences in Monkey B and only minor changes in Monkey A (**Fig. 11H, P**). All statistical comparisons were made against the pre-surgery control period using a two-sample t-test with Bonferroni correction ($\alpha = 0.01$).

Monkey A



Monkey B

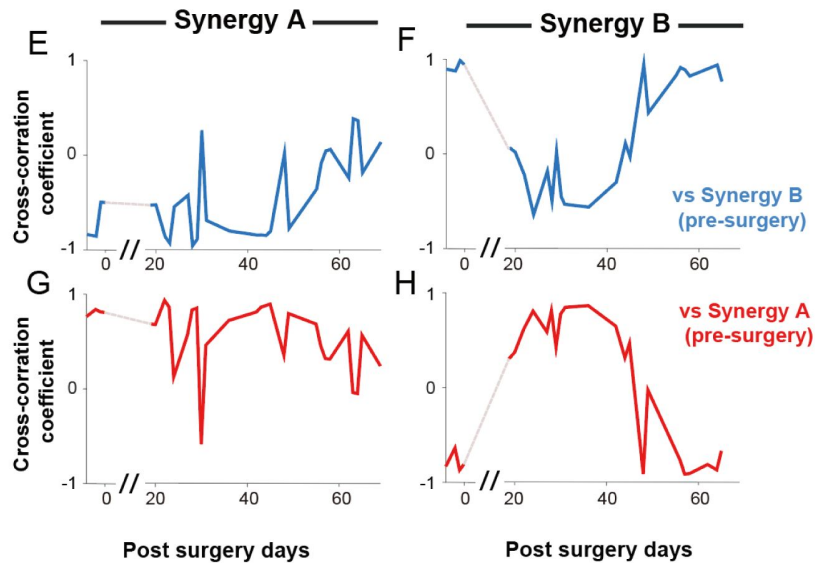
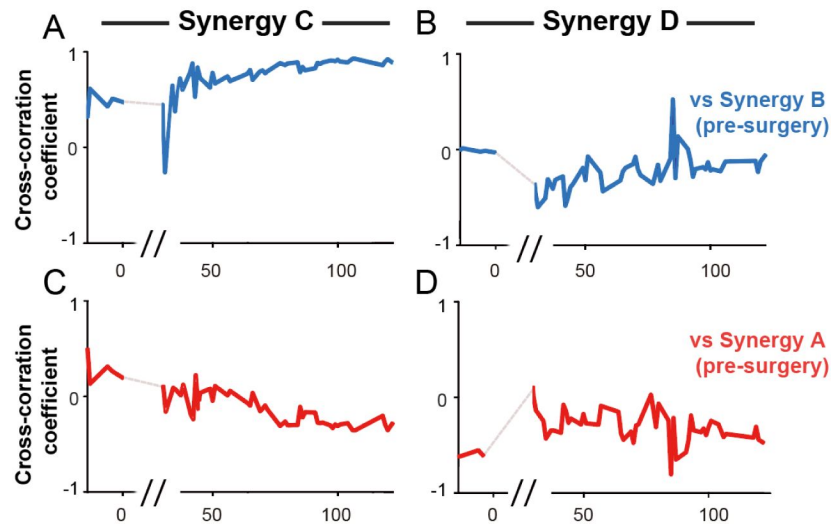


Figure 9

Cross-correlation analysis reveals the two-phase adaptation of primary synergy activation.

Cross-correlation coefficients for the primary flexor (Synergy A) and extensor (Synergy B) synergies are plotted over post-surgery days for Monkey A (A-D) and Monkey B (E-H). For each synergy, the activation pattern was cross-correlated with its own pre-surgery profile (A, B, E, F) and with that of its antagonist (C, D, G, H). The results quantitatively demonstrate the swap-and-revert pattern in both animals. The // represents the recovery period.

Monkey A



Monkey B

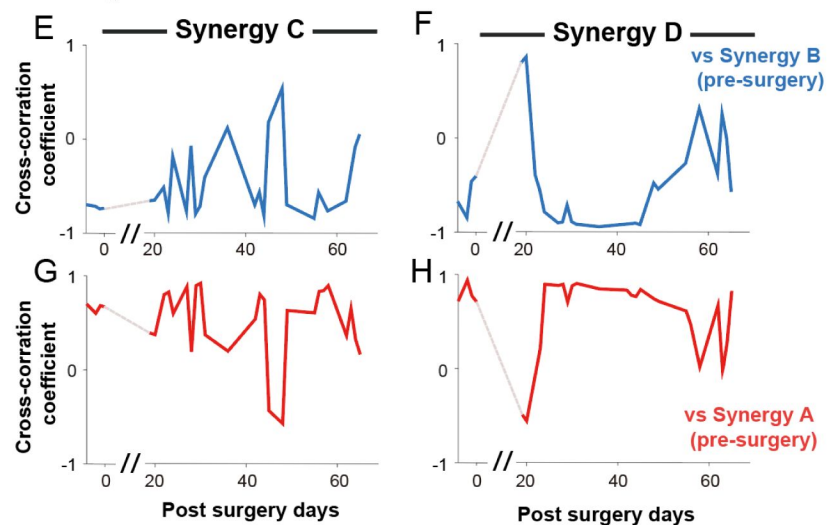


Figure 10

Cross-correlation analysis of secondary synergy activation.

Cross-correlation coefficients for the secondary synergies (C and D) are plotted over post-surgery days for Monkey A (A-D) and Monkey B (E-H). For each synergy, the activation pattern was cross-correlated with the pre-surgery profiles of Synergy A (red lines) and Synergy B (blue lines) to reveal their changing relationships to the primary flexor and extensor commands.

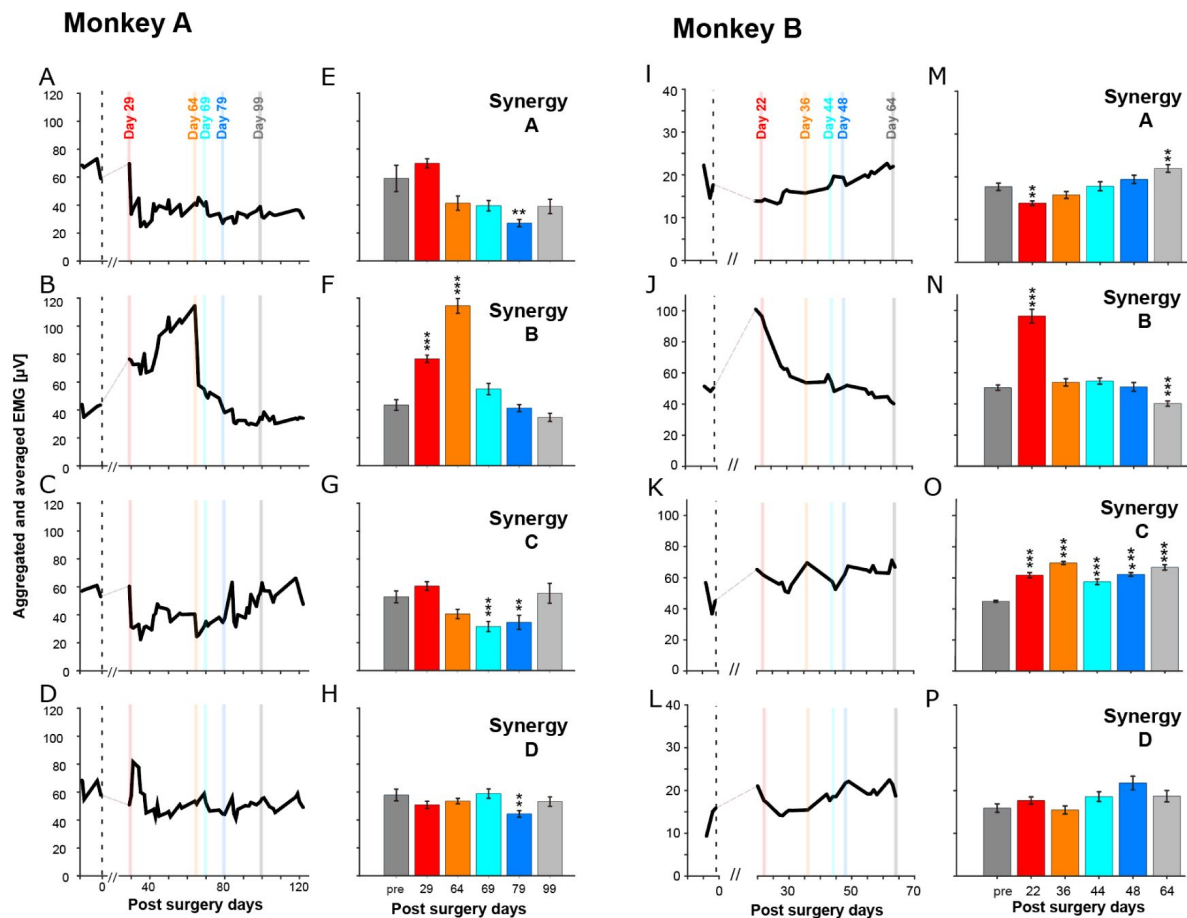


Figure 11

Aggregated and averaged EMG (aaEMG) reveals synergy-specific adaptive patterns.

Aggregated and averaged electromyography (EMG) activities for the main contributing muscles of each of the four synergies for Monkey A (A-H) and Monkey B (I-P). For each recording session, EMG activity within a $\pm 15\%$ task range window was summed for the contributing muscles and then averaged. The left columns for each monkey (A-D, I-L) show the aaEMG over all post-surgery days. The right columns (E-H, M-P) show the mean ($\pm$ SEM) aaEMG for the pre-surgery period ("pre") and the five selected landmark days. Monkey A (A-H) The contributing muscles for each synergy were: Synergy A: flexor digitorum superficialis (FDS) and flexor carpi ulnaris (FCU). Synergy B: extensor digitorum communis (EDC), extensor digitorum-2,3 (ED23), and extensor carpi ulnaris (ECU). Synergy C: palmaris longus (PL) and flexor carpi radialis (FCR). Synergy D: flexor digitorum profundus (FDP), brachioradialis (BRD), and extensor carpi radialis (ECR). Monkey B (I-P) The contributing muscles for each synergy were: Synergy A: flexor digitorum profundus (FDP) and palmaris longus (PL). Synergy B: extensor digitorum communis (EDC), extensor digitorum-2,3 (ED23) and extensor digitorum-4,5 (ED45). Synergy C: extensor carpi radialis (ECR), brachioradialis (BRD), and deltoid (DEL). Synergy D: extensor carpi ulnaris (ECU). Vertical colored bars on the time-series plots indicate the landmark days shown in the bar plots. The // on the time axis represents the recovery period during which no recordings were taken. Asterisks on bar plots indicate a significant difference from the pre-surgery control period ($p < 0.01$, $*p < 0.001$, $**p < 0.0001$; two-sample t-test with Bonferroni correction, $\alpha = 0.01$).

Taken together, these distinct patterns of aggregated EMG activity—especially the escalating, conflicted activation in Synergy B—further illustrate that the early adaptive phase was characterized by an unstable and inefficient neural control strategy.

Recruitment of a Compensatory Tenodesis Strategy Enables Late Adaptation

The development of a compensatory strategy, primarily involving the wrist flexor synergy (Synergy C), appears crucial for enabling the final phase of adaptation. The increasing contribution of this synergy is supported by cross-correlation analysis, which shows that the correlation between Synergy C and the extensor synergy (Synergy B) steadily increased over time after an initial drop (**Fig. 10A**; $p < 0.05$, two-sample t-test). Increased reliance on synergy C following TT suggests a compensatory strategy to overcome the difficulty in finger extension, crucial for grasping. This is despite the recruitment of the finger extensor and wrist flexor synergies, which seemed inadequate to achieve effective flexion. The interactions between the cumulative proximal-to-distal joint angles and musculo-tendon length can be complex. In our experiments, as in human patients who receive tendon transfers (Zajac, 1992), those interactions can affect the ranges of finger flexion-extension that can be achieved for a given wrist angle (i.e., passive and active tenodesis). Briefly, biomechanically, wrist flexion can enable finger extension (i.e., a dynamic tenodesis where extensor tendons of the fingers cross both the wrist and interphalangeal joints): Finger extension can cancel out the increase in total musculo-tendon length produced by wrist flexion. This compensatory interaction across joints allows for some degree of functional hand use, even in the absence of active finger movement (Cash and Jones, 2011) (see the Discussion for a complete description).

Further kinematic analysis of Monkey B (**Fig. 12**; no analysis on Monkey A due to low-resolution video images) indicated that this monkey indeed exhibited a similar active tenodesis as human patients. The angle at the metacarpophalangeal (MCP) joint, responsible for finger extension, steadily increased (**Fig. 12A**) after surgery, indicating finger extension. Concurrently, the wrist joint angle decreased (**Fig. 12B**), indicating wrist flexion. This coordinated movement suggests a compensatory strategy to achieve finger extension during grasping. Statistical analysis confirmed that MCP and wrist joint angles differed significantly between pre- and post-TT ($p < 0.0001$, ANOVA; $\alpha = 0.01$). Therefore, we suggest that deteriorated finger extension after TT was assisted by a compensatory wrist flexion synergy.

Discussion

The central nervous system's (CNS) response to profound musculoskeletal change is a fundamental problem in motor control. This study sought to determine whether the primate CNS adapts to such change by flexibly modulating stable muscle synergies or by developing more fractionated, independent muscle control. We found that the CNS initially defaulted to a modular strategy, repurposing entire synergies by swapping their activation timings. This simple solution, however, proved to be maladaptive, creating a mechanical conflict that impaired motor function. This early maladaptive phase was ultimately resolved through the gradual development of compensatory movements, leading to a “good enough” functional recovery. This multi-stage process, operating on different timescales, highlights the intricate balance between modularity and flexibility in neural adaptation.

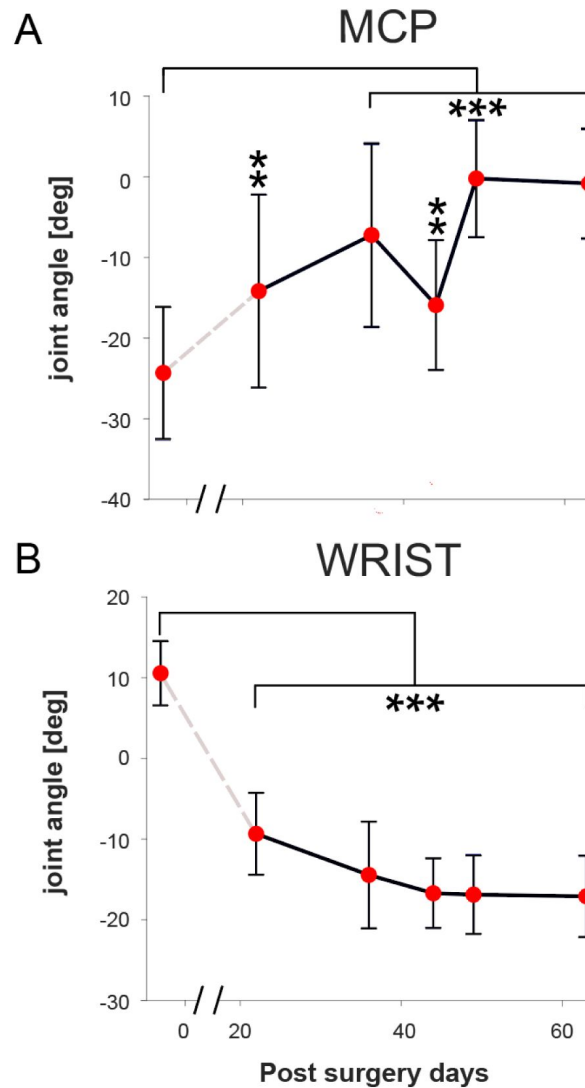


Figure 12

Kinematic analysis reveals a compensatory tenodesis strategy in Monkey B.

Changes in joint angles at the wrist and metacarpophalangeal (MCP) joint for Monkey B (mean of 20 trials $\pm$ SD for each landmark day, taken 83ms before food touch). (A) The MCP joint angle increased post-surgery, indicating greater finger extension. (B) Concurrently, the wrist joint angle decreased, indicating wrist flexion. This coordinated movement pattern is characteristic of an active tenodesis effect, suggesting it was used as a compensatory strategy to achieve finger extension. Statistical analysis confirmed these changes were significant ($p < 0.0001$, $\alpha = 0.01$, ANOVA).

The CNS Defaults to a Modular Strategy, Leading to a Maladaptive Conflict

Our primary finding provides a direct answer to the question posed in the introduction: the CNS adapted to the tendon transfer not by developing fine-grained, fractionated control, but by implementing a modular strategy by repurposing entire co-activation modules (Overduin et al., 2008 [↗](#); Berger et al., 2013 [↗](#); Bizzi and Cheung, 2013 [↗](#); Takei et al., 2017 [↗](#)). The initial and most immediate neural change was a wholesale swap of the temporal activation patterns of the primary flexor and extensor synergies (Synergy A and B). The fact that the muscle weight compositions of all four synergies remained remarkably stable throughout the months-long experiment (Fig. S4), in the face of drastically altered biomechanics, strongly supports the hypothesis that these synergies represent stable, neurally constrained building blocks (d’Avella et al., 2003 [↗](#); Bizzi and Cheung, 2013 [↗](#); Takei et al., 2017 [↗](#)). Our result indicated that this modular approach is likely the default for adaptation to altered musculoskeletal systems because it is computationally “easy” and metabolically “cheap” (Flash and Sejnowski, 2001 [↗](#)). It relies on the simple retiming of pre-existing, stable neural circuits—thought to be encoded in the spinal cord and brainstem—which is a far less demanding operation than the “difficult” alternative of fractionated control (Bizzi and Cheung, 2013 [↗](#)). The latter would require extensive synaptic plasticity and cortical remapping to selectively uncouple muscles that were previously co-activated, a process associated with the slower, more deliberate learning of an entirely new motor skill (Kitago and Krakauer, 2013 [↗](#)). Thus, the simple and economical strategy is prioritized in the CNS to deal with an acute alteration of the musculo-skeletal system.

However, this adherence to established synergy structures created a fundamental conflict. Activating the original extensor synergy (Synergy B) after the transfer now inevitably co-activated the surgically transferred EDC, which functioned mechanically as a flexor, alongside non-transferred muscles like ED23 and ECU, which remained anatomical extensors. This internal mechanical antagonism appears to be the root cause of the early maladaptive phase. This interpretation is supported by two key lines of evidence from our results. First, the period of severe behavioral impairment, characterized by aberrant reaching and inefficient grasping (Fig. 5B [↗](#)), occurred precisely when this flawed “swap” strategy was active. Second, the aggregated EMG activity revealed a sustained and significant increase in the total activation of the conflicted Synergy B in Monkey A (Fig. 11B, F [↗](#)), likely reflecting an energetically costly effort to overcome the internal mechanical antagonism. This scenario can be viewed through the lens of optimal control and cost-benefit analysis, where a cost function is minimized (Wolpert, 1997 [↗](#)). The initial “swap” strategy, while computationally cheap to select, incurred an unacceptably high operational cost in terms of both poor task performance (high error) and excessive energy expenditure. It was this profoundly unfavorable cost-benefit ratio that likely served as the critical error signal that drove the CNS to abandon this initial strategy.

Resolution Through Slower Compensatory Adaptations

The CNS did not persist in the failed swap-based strategy. The high metabolic and computational cost of activating a mechanically conflicted synergy likely triggered the second rapid adaptation: the “switch-back” of synergy activation timings toward their original patterns (Fig. 7 [↗](#), 9 [↗](#)). This rapid reversion, occurring over just a few days, is characteristic of an error-based learning mechanism, a form of adaptation that is profoundly impaired by cerebellar damage across a range of tasks, including adaptation to force fields (Smith and Shadmehr, 2005 [↗](#)), prismatic shifts (Martin et al., 1996 [↗](#)), and split-belt treadmills (Morton and Bastian, 2006 [↗](#)). It is plausible that the CNS operates with an implicit threshold for an acceptable cost/performance ratio; once the persistent task failure and high muscular co-contraction of the swap strategy exceeded this threshold, a swift recalibration was initiated. This reversion, despite the remaining mechanical antagonism from the transferred tendons, represents a “good enough” solution where functional success is prioritized over perfect efficiency (Mussa-Ivaldi and Bizzi, 1995 [↗](#); Ranganathan et al.,

2013 [↗](#); Gijssberts et al., 2014 [↗](#)). The rapid timescale of this change is highly consistent with the cerebellum's proposed role as a forward model, predicting the sensory consequences of motor commands and driving rapid learning in response to sensory prediction errors (Popa et al., 2016 [↗](#)).

Crucially, this switch-back to a less conflicted state, presumably representing the involvement of the error-based learning process, was likely enabled only by the concurrent development of slower, compensatory strategies that provided an alternative means to achieve the task goal. The primary compensation was a learned use of the tenodesis effect. Over weeks, the monkeys gradually increased the activation of the wrist flexor synergy (Synergy C) during hand opening. This wrist flexion biomechanically generates passive finger extension (Zajac, 1992 [↗](#); Cash and Jones, 2011 [↗](#)), providing a viable new method for hand pre-shaping. This learned, compensatory behavior, confirmed by kinematic analysis (Fig. 12 [↗](#)), ultimately allowed the CNS to abandon the maladaptive synergy swap. This gradual, exploratory process is distinct from the following rapid adaptation and represents a form of motor skill acquisition, a process which may be associated with plasticity in cortical structures like the motor cortex and basal ganglia. (Kitago and Krakauer, 2013 [↗](#)).

This multi-timescale process, conceptually summarized in Figure 13 [↗](#), aligns well with influential two-state models of motor learning, which propose parallel “fast” and “slow” learning systems (Smith et al., 2006a [↗](#)). These models posit that a fast system responds strongly to error but has poor retention, while a slow system responds weakly to error but retains information well, enabling robust long-term memory. The initial synergy swap and subsequent “switch-back” appear to be governed by the fast system (likely cerebellar), which responds strongly to error but has poor retention. The gradual learning of the tenodesis effect is the signature of the slow system (likely cortical and/or basal ganglia-dependent), which integrates experience over a longer period to build robust, lasting skills (Trewartha et al., 2014 [↗](#)). The interaction between these systems is key: as the slow system gradually develops the effective tenodesis compensation, it reduces the overall task error. This reduction in error removes the primary drive for the fast, error-correcting system to continue its search for a solution. With its error signal quieted, the fast system allows the control policy to stabilize, and the CNS “settles” into the “good enough” compensatory state. This suggests a control architecture where the successful output of a slow, deliberate skill-learning system can gate and ultimately supersede the output of a fast, reactive adaptation system, leading to a stable and functional long-term motor plan.

Broader Implications and Limitations

Our findings reveal a complex interaction between adaptive processes operating on different timescales. The initial, rapid synergy swap may be linked to a ‘sensory surprise’ from altered proprioceptive feedback (Kitazawa et al., 1995 [↗](#); Martin et al., 1996 [↗](#); Smith et al., 2006b [↗](#); Edwards et al., 2012 [↗](#); Petit et al., 2018 [↗](#)), driving a fast but flawed response. This multi-level network reorganization presents a more complex challenge than the gain modulation within single or specific CNS pathways seen in studies of spinal reflex conditioning (Thompson and Wolpaw, 2014 [↗](#)). Further understanding the interaction between the global synergy reorganization observed in this study and the underlying local sensorimotor loops will require considering factors not assessed here, such as the roles of spinal reflex pathways (Smith et al., 2006c [↗](#)) and fusimotor drive (Vallbo et al., 1979 [↗](#); Hospod et al., 2007 [↗](#)). The altered biomechanics would have immediately changed proprioceptive feedback, triggering rapid modulation of spinal reflex circuits and requiring recalibration of the fusimotor system, which adjusts muscle spindle sensitivity via alpha-gamma co-activation. A comprehensive model must ultimately connect the global synergy reorganization with these local, interacting feedback and control loops.

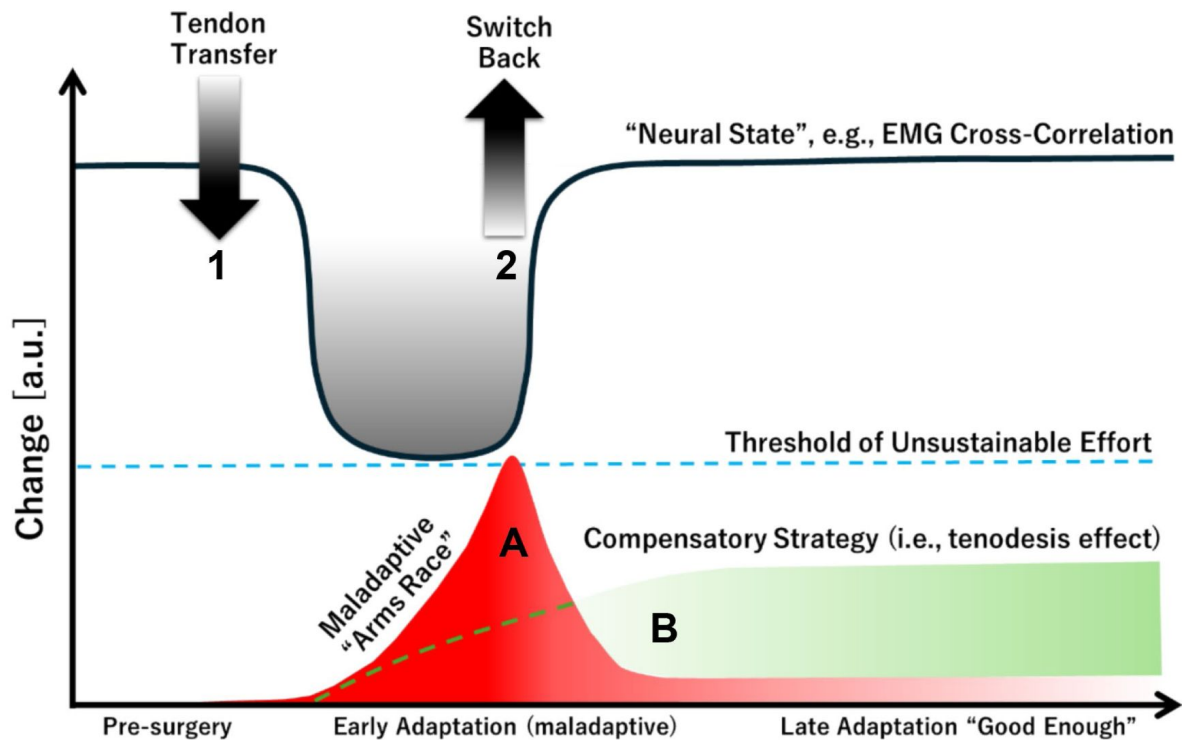


Figure 13

A Proposed Model of Multi-Timescale Adaptation Following Tendon Transfer.

This schematic illustrates the hypothesized interaction between fast and slow adaptive processes driving recovery. The initial Tendon Transfer triggers a rapid but maladaptive ‘swap’ of motor commands (Fast Adaptation 1), leading to a maladaptive state. During this phase, two slower processes are hypothesized to occur in parallel: a costly ‘arms race’ within the conflicted synergy (Slow Process A, red curve) and the gradual development of a functional compensatory strategy (Slow Process B, green curve). When the ‘arms race’ reaches a threshold of unsustainable cost (dashed blue line), a second Fast Adaptation (‘Switch-Back’, 2) is triggered. This allows for the abandonment of the flawed strategy and the adoption of a stable, ‘good enough’ solution, which is now supported by the newly learned compensatory strategy. The gray line represents the observed neural data (e.g., cross-correlation coefficients of electromyography and temporal activation coefficients of muscle synergies), which reflects this two-phase process.

While our primary analysis focused on the grasping, the adaptation encompassed the entire reach-to-grasp sequence, as evidenced by the initial deficits in reaching movements (Supplementary Videos S2, S3, S6; **Fig. 5B**). The observation that both monkeys, engaged in both a highly controlled learned task and a more naturalistic grasping task, showed similar adaptive patterns strengthens the generalizability and ecological validity of our findings (Berniker et al., 2013). Furthermore, the qualitative reports of proficient real-world hand use indicate that the learned compensatory strategies were not narrow “tricks” but robust solutions that transferred to unconstrained activities, demonstrating their “portability” to the complexities of natural behavior.

However, a limitation is that this generalization was not tested systematically. Another critical point is that our paradigm cannot distinguish between the unlearning of a maladaptive strategy and the relearning of a functional one. This distinction is crucial for both motor memory theory and clinical practice (Kitago et al., 2013). The disappearance of the initial “swap” strategy could be due to passive decay (forgetting), active inhibition of the memory (extinction), or the formation of a new, stronger memory that outcompetes it (interference) (Villalta et al., 2013). The literature increasingly suggests that unlearning is an active process, and that the new, successful compensatory strategy likely supplants the maladaptive one through interference. This question is vital for rehabilitation: should therapy focus on “erasing” bad habits, or on intensively training better ones to overwrite them? Our results lend weight to the latter, suggesting functional recovery is a process of competitive memory formation.

Implications for Sensorimotor Rehabilitation

These findings may hold implications for sensorimotor rehabilitation. The observation of a distinct maladaptive phase followed by compensation and recovery suggests a staged approach could be beneficial (Hatem et al., 2016; Kakavas et al., 2025). An early phase of therapy might focus on minimizing sensory prediction errors caused by the altered proprioceptive feedback, and reducing maladaptive responses through techniques aimed at integrating the altered body schema (e.g., using visual feedback or task-oriented practice) (Nudo et al., 1996; Takeuchi and Izumi, 2012; Zeiler and Krakauer, 2013; Foell et al., 2014; Levin et al., 2015; De Nunzio et al., 2018; Sattin et al., 2023; van Vliet et al., 2023). This stage would guide the fast, error-driven learning system. A later phase could then employ task-specific training to promote effective compensatory strategies, facilitating the abandonment of maladaptive patterns and the consolidation of a stable motor strategy (Murata et al., 2008; Maier et al., 2019). This second stage engages the slow, skill-acquisition system by creating a rich, problem-solving environment. Finally, a third stage would focus on intensive, high-repetition practice to automate the new functional motor plan, making it robust for real-world use through dosage and generalization (Winterbottom and Nilsen, 2024). This neurobiologically-informed staging—guiding fast adaptation, then facilitating slow skill learning, then driving consolidation—may offer a more logical and effective path to functional recovery.

Methods

Animals

Data were collected using two male macaque monkeys (*Macaca fuscata*; Monkey A: 7.8 kg and Monkey B: 9.9 kg) purpose-bred at the National Bioresource Project (NBRP). They were kept in custom-made primate cages, allowing for potential pair housing. Both monkeys were trained to perform a simple grasping task. During experimental recordings, the monkeys were seated in a primate chair without further restraint. Their head movement remained unrestricted at all times.

Procedures

All procedures were designed to minimize discomfort and pain and approved by the institutional Animal Care and Use Committees at the National Center of Neurology and Psychiatry (NCNP), Tokyo, Japan. Details of the surgical operations, experimental setup, and procedures for electromyography (EMG) recordings have been previously described (Takei and Seki, 2010 [↗](#)).

EMG implant surgery

Both animals were familiarized with the experimental setup and trained to perform the behavioral task prior to surgery. After an initial training period, EMG electrodes were chronically implanted subcutaneously into several muscles of the left forearm including the FDS, FDP, and EDC (Figs. 1 [↗](#), 3C [↗](#), and 4C [↗](#)). Individual muscles were localized and confirmed using electrical microstimulation before implanting EMG electrodes chronically by two methods. For muscles involved in the crossed tendon transfer procedure, muscle fascia was cut and EMG wires with looped ends were placed on top of the muscle belly. These were then secured by reclosing the fascia with absorbable suture threaded through the loop (AS-361, stainless-steel Cooner wire, Conner Wire Co., Chatsworth, CA, USA). All other muscles were directly implanted into the muscle belly using a 22-gauge hypodermic needle. Each wire was threaded into the needle tip, folded back along the shaft, and inserted into the muscle before carefully retracting the needle. Electrode separation for bipolar recordings was approximately 5–10 mm. Note that some EMG recording sites were lost over time. Surgical procedures were carried out under deep general anesthesia (sevoflurane 1.5 to 2.5% in 2:1 O₂/N₂O) and with full aseptic precautions. Heart rate, blood pressure, body temperature, and blood oxygen saturation were monitored throughout surgery. Analgesics and antibiotics were administered intramuscularly for at least one week postoperatively.

Tendon-Transfer Surgery

After a minimum four-week recovery period from EMG implant surgery, TT surgery was performed. The FDS and EDC tendons were cut as distally as possible (immediately below and above the wrist for FDS and EDC, respectively) to avoid damaging the Golgi tendon organ located near the junction between the muscle fibers and tendon (Schoultz and Swett, 1972 [↗](#); Jami, 1992 [↗](#)). Tendons were then guided either through the gap between the radius and ulnar bone (Monkey A), or around the wrist and then re-attached to the tendon of the antagonist muscle using a tendon graft harvested from the plantaris tendon of the lower limb (Monkey B; EDC → FDS: direct connection; and FDS → EDC: tendon grafting). This was only necessary in Monkey B since the tendons used for the cross-transfer were too short. This surgical procedure aimed to reverse the primary mechanical actions of the manipulated muscles, making the FDS tendon an effective finger extensor and the EDC tendon an effective finger flexor. The success and nature of this mechanical rearrangement were verified post-operatively via direct muscle stimulation (see Methods subsection ‘Tendon transfer confirmation’ and Fig. 2 [↗](#)). The general organization and function of the major forearm muscles involved in finger flexion and extension in *Macaca fuscata* are broadly comparable to those in humans (Vanhoof et al., 2021 [↗](#); Yan et al., 2022 [↗](#)).

Surgical procedures were carried out under deep general anesthesia (sevoflurane 1.5 to 2.5% in 2:1 O₂/N₂O) and with full aseptic precautions. Heart rate, blood pressure, body temperature, and blood oxygen saturation were monitored throughout surgery. Analgesics and antibiotics were administered intramuscularly for at least one week postoperatively. Both monkeys wore a plaster cast post-surgery, but its effectiveness was limited (lasting approximately one week).

Tendon transfer confirmation

To verify the success and long-term stability of TT surgery, two procedures (**Fig. 2**) were implemented. During the first procedure, the monkey was sedated and its arm was secured to a metal frame (**Fig. 2A**). Electrical stimulation was then applied to either the FDS or EDC muscle (50 mA, DS8R, Digitimer, Welwyn Garden City, UK), while ultrasound scans of tendon movements were concurrently captured (SONIMAGE MX1, Konica Minolta, Inc., Tokyo, Japan (Nordez et al., 2009; Dieterich et al., 2017)).

Figure 2B shows a sonogram of the FDS muscle and its intramuscular tendons. The left side displays a static image of the FDS muscle at a specific moment, with white arrows indicating the FDS tendon used for measurement. The right side shows staggered images (top white box) of the FDS tendon displacement triggered by muscle stimulation. The lower right inset illustrates the measurement area. The displacement wave area represents the intensity of muscle contraction, and was calculated by measuring the average duration (a, in seconds) and amplitude (b, in cm) of three successive waves. The area ($a \cdot b / 2$) and regression line for FDS (red) and EDC (blue) ($R^2 > 0.5$ for FDS) for days 0, 7, and 105 post-TT are shown (**Fig. 2C**). The results suggest that the muscle contractions induced by direct electrical stimulation remained nearly constant.

For the second procedure, high-speed videos (37U Series Color Industrial Camera, The Imaging Source, Charlotte, NC, USA) of the monkey's finger movements were recorded while electrically stimulating the FDS or EDC. Markers were placed on the nails of the index, middle, and ring fingers (**Fig. 2A**) to measure finger displacement in xyz-dimensions using DeepLabCut software (Mathis et al., 2018; Nath et al., 2019). The sum of Euclidean distances of each marker from the origin of the three-dimensional (3D) coordinate system was computed as a scalar quantity. Post-surgery, movement along the z-axis reversed, indicating a shift from finger flexion to extension due to TT (**Fig. 2D**; blue: pre-TT on surgery day, dark brown: post-TT on surgery day, light brown: 1 wk post-TT, and red: 3 wk post-TT). The scalar quantities of each finger did not significantly change during muscle stimulation on days 0-, 7-, and 105-days post-TT (**Fig. 2E**), suggesting there was no postoperative tendon rupture or slackening.

Data recordings

EMG signals were recorded using the AlphaLab SnR system (Alpha Omega Engineering, Hamerkava St. 6, Ziporit Industrial Zone, P.O. Box 810, Nof HaGalil (Nazareth Illit) 1789062, Israel), displayed online and then stored on a hard drive for later off-line analysis using MATLAB (MathWorks, Natick, MA, USA). Data were recorded at 11 kHz and processed for analysis as follows. 1. Down-sampled to 5 kHz; 2. 50 Hz high-pass filtered (6th order Butterworth filter); 3. rectified; 4. 20 Hz low-pass filtered (6th order Butterworth filter); and 5. down-sampled to 100 Hz. Behavioral task events were recorded as transistor–transistor logic (TTL) signals using the AlphaLab SnR system. Monkey behavior was further recorded by two cameras (Sanyo VPC-WH1, 60 fps [Sanyo, Osaka, Japan]; and Casio EX-100F, 240 fps [Casio, Tokyo, Japan] for monkeys A and B, respectively) from two different angles (top and side view). The images were later used to detect additional behavioral events (contact time with the object and pull onset for Monkey A; and food touch and food lift onset for Monkey B), and detection of possible changes in the animal's general movement pattern. Kinovea software was used for this video analysis (free and open-source software; <https://www.kinovea.org/>).

Behavioral task

Before and after EMG surgery, the monkeys were trained on a simple grasping task that involved a small object attached to a rod (**Fig. 3B** for Monkey A; and **Fig. 4B** for Monkey B). For Monkey A, object 1 was a small rod placed between two side walls encouraging the monkey to grasp the rod with a precision grip using the tips of the index finger and thumb (controlled grasp). The force

used to compress the spring while pulling was low and the distance moved short. Object 2 had to be grasped in the same way. However, the force to compress the spring was higher and the distance moved longer (object 1: 300 cN, 4 mm; and object 2: 800 cN, 30 mm). The monkey was asked to grasp and hold the objects for approximately 300–500 ms (**Fig. 3A–D**). After completing the trial, the monkey was rewarded with a piece of fruit that had to be taken from the experimenter's palm. The task sequence for Monkey A ranged from object 1 hold onset to object 2 hold offset (**Fig. 3C, D**; and 'Data Analysis' below). For Monkey B, object 1 was a small rod that was grasped and pulled (300 cN, 4 mm) using a lateral prehension grip (like using a door key), and resembling a power grip rather than a natural grasp. However, instead of grasping object 2, Monkey B picked up a piece of fruit from an allocated location. In doing so, the monkey had to cross the path of a photo cell which was detected by the recording system (**Fig. 4A–D**). The task sequence to be analyzed for Monkey B ranged from object 1 hold onset to food touch, which was indicated by a LED (**Fig. 4C, D**; and 'Data Analysis' below).

For Monkey A, the task sequence was as follows (**Fig. 3A**; and Supplementary Video S1). From a starting position in front of the monkey, it had to lift its arm and move towards object 1, grasp, pull, and then hold the object for 500 ms. Immediately after releasing object 1, Monkey A proceeded to move its hand towards object 2, which was located to the right of object 1. Again, the monkey had to grasp, pull, and then hold the object for another 500 ms. Each hold period was accompanied by an audio signal which stopped once the hold duration was sufficient. After completing the task, which was signaled by another acoustic signal, the monkey was rewarded with a piece of fruit presented by the experimenter and taken by the monkey.

For Monkey B, the first part of the sequence was identical. However, after releasing object 1, Monkey B was required to pick up a piece of fruit from an allocated location (**Fig. 4A, B**; and Supplementary Video S5). In doing so, the monkey passed a photocell in front of the food well. This event was detected by the recording system and a video camera (a red LED triggered by activation of the photocell). For the analysis, only object 1 and the food grasp were considered in monkeys A and B, respectively.

Data analysis

EMG analysis

EMG data were normalized to the average time for each monkey to complete a trial (object 1 hold onset → object 2 hold offset/LED offset = 100%). Data were cropped and aligned according to the time stamps (Obj1 hold offset ± 15%; and LED onset [food touch] ± 15% for monkeys A and B, respectively). Recorded EMGs from the pre- and post-tendon surgery period were averaged for each recording and compared over experimental sessions. For both monkeys, the muscles used for EMG analysis were: FDS and EDC for pre-(control) and EDC for post-TT (experimental).

Both monkeys required time to recover from surgery. Once they were able to perform the task on their own, data collection started (29 and 20 days post-surgery for monkeys A and B, respectively). After tendon surgery, EMG signals for FDS, FCU, and FCR deteriorated in Monkey B. Therefore, there is no experimental data for these muscles in this monkey and they were excluded from the muscle synergy analysis.

Synergy analysis

The EMG envelopes obtained from pre-processing raw EMG data were divided by the mean value to normalize activity. Muscle synergies were then extracted for each session using NMF (d'Avella et al., 2003). NMF decomposes the EMG data matrix M , as a product of two matrices C and W :

$$M = CW^T + re = \sum c_i w_i^T + re$$

where the vector w_i and column of matrix W are muscle synergies; the vector c_i , and column of matrix C are their temporal coefficients; and re represents the residuals. Multiplicative update rules were used for decomposition (Lee and Seung, 1999). Updating matrices for decomposition from 20 different random initial matrices was started, and factorization results with least residuals among the 20 results were used. To improve robustness of the muscle synergies, one set of muscle synergies was extracted from EMG data of multiple trials of each day. A k -fold cross validation was used, with $k = 4$ i.e., EMG data set of multiple trials during the day was split into four data sets, in which three data sets were for training and one data set was for testing. Synergy matrix W was calculated from three training data sets using NMF, and then a coefficient matrix C was calculated by decomposing the EMG matrix M of the test data set with a fixed W matrix of the training data set. This operation was repeated by changing the test data set for each four data sets. Finally, the extracted four C matrices and four W matrices were averaged to obtain daily synergies.

The number of synergies was determined as the number where VAF exceeded a threshold. This threshold was set as 0.8. To clarify the effect of TT, the number of synergies for each monkey was determined, assuming that trial-by-trial variation in synergies within the same monkey was small. MATLAB 'nnmf' function was used for NMF.

To determine whether the weights of muscle synergies changed after TT, the following procedure was used. First, the average of each pre-TT spatial synergy was calculated and used as control data to calculate the cosine distance for all post-TT spatial synergies. This generated four distance relationships for each pre-TT synergy. If the change in spatial synergy before and after TT was small, then only one of the four cosine distances (e.g., synergy A before TT and synergy A after TT) should be significantly smaller. Two-way ANOVA was conducted using the type of synergy and session as factors for the cosine distance. The significance of synergy pairs was identified using the Bonferroni post-hoc test.

Cross-correlation analysis

Cross-correlation analysis was performed to examine pre- and post-surgery EMG signals as well as temporal activation coefficients of extracted muscle synergies. Cross-correlation coefficients were calculated using MATLAB (MathWorks, `corrcoef.m` function). These were plotted for different muscles and muscle synergy activation coefficients over experimental sessions for comparison.

Based on cross-correlation analysis, five 'landmark days' were identified for further analysis (**Fig. 6P**). These days represent distinct stages in the recovery process and excluded pre-surgery control data. The first landmark day was chosen from one of the initial recording sessions after TT surgery (days 29 and 22 in monkeys A and B, respectively) when the cross-correlation coefficients had changed significantly compared with control data. The second day was then chosen from a time period just before the switch-back, when the cross-correlation coefficients had started to return to their original values (days 64 and 36). Another day was then picked from the period when the coefficients were still changing significantly (days 69 and 44) and before starting to saturate. At this point the next day was defined (days 79 and 48). Finally, the final landmark day was chosen from one of the last recording sessions when the behavior had fully recovered (days 99 and day 64 for monkeys A and B, respectively).

Behavioral analysis

To examine behavioral recovery, the duration, onset, and offset of object and food grasps were analyzed. Event times extracted from the video analysis were used for alignment of EMG data and subsequent cross-correlation and synergy analyses, along with recorded event TTL signals. For each experimental session, the first 20 trials of each video recording were analyzed. In Monkey A, the following events were detected and in-between times stored in ms: touch onset, pull onset, and hold onset (resulting in grasp duration and pull time) for object 1 and object 2. In Monkey B, the

following events were detected and in-between times stored: touch onset, pull onset, and hold onset (resulting in grasp duration and pull time) for object 1; and for the food grasp component of the task, LED onset, food-touch onset, food-lift onset, LED offset, and movement end (contact of food with mouth).

Contact times (touch onset → pull onset) for Monkey A were plotted in ms (mean ± SD) over number of days from tendon surgery for object 1. For Monkey B, contact times with food (food-touch → food-lift onset) were plotted in ms (mean ± SD).

ANOVA was performed to compare control data recorded before and after tendon surgery. In total, 5 and 3 control sessions vs. 35 and 22 experimental sessions were used for analyses in monkeys A and B, respectively.

To quantify the observed aberrant reaching behavior, video footage was analyzed and the means of 10 consecutive trials calculated for each session as follows. For Monkey A, time spent within a specific spatial window covering the area behind and between the two objects was measured; the monkey passed through this space almost exclusively while exhibiting the impairment (see **Fig. 3E**, crossing of the yellow dotted line). In Monkey B, the time spent in contact with the rear plate of the object was measured (see **Fig. 4G**).

Joint kinematics analysis

To determine whether compensatory movements occurred, steady state changes in joint angles were examined before and after recovery. Using DeepLabCut (Mathis et al., 2018; Nath et al., 2019), key points on the fingers and wrists were tracked from experimental videos. From the obtained key points, the vector v_1 connecting the wrist and fingers and vector v_2 connecting the arm and wrist were estimated. The joint angle θ was calculated as follows:

$$\cos(\theta) = \frac{v_2 \cdot v_3}{|v_2| |v_3|} \rightarrow \theta = \cos^{-1} \left(\frac{v_2 \cdot v_3}{|v_2| |v_3|} \right)$$

Data, materials, and software availability

Matlab scripts and raw data will be made available online. All study data are included in the article and/or SI Appendix.

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Additional information

Author contributions

R.P. and K.S. designed all experiments and wrote the paper. R.P., Y.H., N.U., and N.O. performed the experiments and analyzed the data. R.P., Y.H., T.O., and K.S. performed the surgeries. T.F. provided conceptual advice and wrote the paper. All authors reviewed the results and approved the final version of the manuscript.

Additional files

supplementary informations [↗](#)

Video 1 [↗](#)

Video 2 [↗](#)

Video 3 [↗](#)

Video 4 [↗](#)

Video 5 [↗](#)

Video 6 [↗](#)

Video 7 [↗](#)

References

Berger DJ, Gentner R, Edmunds T, Pai DK, d'Avella A (2013) **Differences in Adaptation Rates after Virtual Surgeries Provide Direct Evidence for Modularity** *The Journal of Neuroscience* **33**:12384–12394 [Google Scholar](#)

Berniker M, O'Brien MK, Kording KP, Ahmed AA (2013) **An Examination of the Generalizability of Motor Costs** *PLOS One* **8**:e53759 [Google Scholar](#)

Bizzi E, Cheung VC (2013) **The neural origin of muscle synergies** *Frontiers in Computational Neuroscience* **7** [Google Scholar](#)

- Bowlus TH, Lane RD, Stojic AS, Johnston M, Pluto CP, Chan M, Chiaia NL, Rhoades RW (2003) **Comparison of reorganization of the somatosensory system in rats that sustained forelimb removal as neonates and as adults** *J Comp Neurol* **465**:335–348 [Google Scholar](#)
- Cash DJW, Jones JWM (2011) **The role of tenodesis in surgery of the upper limb** *The Journal of Bone & Joint Surgery British* **93**:285–292 [Google Scholar](#)
- Cheung VC, Turolla A, Agostini M, Silvoni S, Bennis C, Kasi P, Paganoni S, Bonato P, Bizzi E (2012) **Muscle synergy patterns as physiological markers of motor cortical damage** *Proc Natl Acad Sci U S A* **109**:14652–14656 [Google Scholar](#)
- d’Avella A, Saltiel P, Bizzi E (2003) **Combinations of muscle synergies in the construction of a natural motor behavior** *Nature Neuroscience* **6**:300–308 [Google Scholar](#)
- Davidson PR, Wolpert DM (2003) **Motor learning and prediction in a variable environment** *Curr Opin Neurobiol* **13**:232–237 [Google Scholar](#)
- De Nunzio AM, Schweisfurth MA, Ge N, Falla D, Hahne J, Gödecke K, Petzke F, Siebertz M, Dechent P, Weiss T, Flor H, Graimann B, Aszmann OC, Farina D (2018) **Relieving phantom limb pain with multimodal sensory-motor training** *J Neural Eng* **15**:066022 [Google Scholar](#)
- Dieterich AV, Botter A, Vieira TM, Peolsson A, Petzke F, Davey P, Falla D (2017) **Spatial variation and inconsistency between estimates of onset of muscle activation from EMG and ultrasound** *Sci Rep* **7**:42011 [Google Scholar](#)
- Edwards MJ, Adams RA, Brown H, Pareés I, Friston KJ (2012) **A Bayesian account of ‘hysteria’** *Brain* **135**:3495–3512 [Google Scholar](#)
- Flash T, Sejnowski TJ (2001) **Computational approaches to motor control** *Curr Opin Neurobiol* **11**:655–662 [Google Scholar](#)
- Fleury L, Prablanc C, Priot AE (2019) **Do prism and other adaptation paradigms really measure the same processes?** *Cortex* **119**:480–496 [Google Scholar](#)
- Foell J, Bekrater-Bodmann R, Diers M, Flor H (2014) **Mirror therapy for phantom limb pain: brain changes and the role of body representation** *Eur J Pain* **18**:729–739 [Google Scholar](#)
- Gaetz W, Dockstader C, Furlong PL, Amaral S, Vossough A, Schwartz ES, Roberts TPL, Scott Levin L (2023) **Somatosensory and motor representations following bilateral transplants of the hands: A 6-year longitudinal case report on the first pediatric bilateral hand transplant patient** *Brain Res* **1804**:148262 [Google Scholar](#)
- Gardenier J, Garg R, Mudgal C (2020) **Upper Extremity Tendon Transfers: A Brief Review of History, Common Applications, and Technical Tips** *Indian J Plast Surg* **53**:177–190 [Google Scholar](#)
- Gijsberts A, Bohra R, Sierra González D, Werner A, Nowak M, Caputo B, Roa MA, Castellini C (2014) **Stable myoelectric control of a hand prosthesis using non-linear incremental learning** *Front Neurobot* **8**:8 [Google Scholar](#)
- Green HJ (1997) **Mechanisms of muscle fatigue in intense exercise** *J Sports Sci* **15**:247–256 [Google Scholar](#)

- Hatem SM, Saussez G, Della Faille M, Prist V, Zhang X, Dispa D, Bleyenheuft Y (2016) **Rehabilitation of Motor Function after Stroke: A Multiple Systematic Review Focused on Techniques to Stimulate Upper Extremity Recovery** *Front Hum Neurosci* **10**:442 [Google Scholar](#)
- Hoffmann G, Kamper DG, Kahn JH, Rymer WZ, Schmit BD (2009) **Modulation of stretch reflexes of the finger flexors by sensory feedback from the proximal upper limb poststroke** *J Neurophysiol* **102**:1420–1429 [Google Scholar](#)
- Hospod V, Aimonetti J-M, Roll J-P, Ribot-Ciscar E (2007) **Changes in Human Muscle Spindle Sensitivity during a Proprioceptive Attention Task** *The Journal of Neuroscience* **27**:5172–5178 [Google Scholar](#)
- Hunter DJ, Eckstein F (2009) **Exercise and osteoarthritis** *J Anat* **214**:197–207 [Google Scholar](#)
- Jami L (1992) **Golgi tendon organs in mammalian skeletal muscle: functional properties and central actions** *Physiol Rev* **72**:623–666 [Google Scholar](#)
- Kakavas G, Brilakis E, Papatzikou M, Malliaropoulos N, Mazeas J, Forelli F (2025) **Reverse Linear Neuro Periodization Model for Rehabilitation After Arthroscopic Rotator Cuff Repair: A Narrative Review** *Clinics and Practice* **15**:105 [Google Scholar](#)
- Kitago T, Krakauer JW (2013) **Motor learning principles for neurorehabilitation** *Handb Clin Neurol* **110**:93–103 [Google Scholar](#)
- Kitago T, Ryan SL, Mazzoni P, Krakauer JW, Haith AM (2013) **Unlearning versus savings in visuomotor adaptation: comparing effects of washout, passage of time, and removal of errors on motor memory** *Front Hum Neurosci* **7**:307 [Google Scholar](#)
- Kitazawa S, Kohno T, Uka T (1995) **Effects of delayed visual information on the rate and amount of prism adaptation in the human** *J Neurosci* **15**:7644–7652 [Google Scholar](#)
- Lee DD, Seung HS (1999) **Learning the parts of objects by non-negative matrix factorization** *Nature* **401**:788–791 [Google Scholar](#)
- Levin MF, Weiss PL, Keshner EA (2015) **Emergence of virtual reality as a tool for upper limb rehabilitation: incorporation of motor control and motor learning principles** *Phys Ther* **95**:415–425 [Google Scholar](#)
- Loeb GE (1999) **Asymmetry of hindlimb muscle activity and cutaneous reflexes after tendon transfers in kittens** *J Neurophysiol* **82**:3392–3405 [Google Scholar](#)
- Luauté J, Schwartz S, Rossetti Y, Spiridon M, Rode G, Boisson D, Vuilleumier P (2009) **Dynamic changes in brain activity during prism adaptation** *J Neurosci* **29**:169–178 [Google Scholar](#)
- Maier M, Ballester BR, Verschure PFMJ (2019) **Principles of Neurorehabilitation After Stroke Based on Motor Learning and Brain Plasticity Mechanisms** *Frontiers in Systems Neuroscience* **13** [Google Scholar](#)
- Martin TA, Keating JG, Goodkin HP, Bastian AJ, Thach WT (1996) **Throwing while looking through prisms. II. Specificity and storage of multiple gaze-throw calibrations** *Brain* **119**:1199–1211 [Google Scholar](#)

- Mathis A, Mamidanna P, Cury KM, Abe T, Murthy VN, Mathis MW, Bethge M (2018) **DeepLabCut: markerless pose estimation of user-defined body parts with deep learning** *Nature Neuroscience* **21**:1281–1289 [Google Scholar](#)
- Mercuri E, Muntoni F (2013) **Muscular dystrophies** *Lancet* **381**:845–860 [Google Scholar](#)
- Morton SM, Bastian AJ (2006) **Cerebellar contributions to locomotor adaptations during splitbelt treadmill walking** *J Neurosci* **26**:9107–9116 [Google Scholar](#)
- Murata Y, Higo N, Oishi T, Yamashita A, Matsuda K, Hayashi M, Yamane S (2008) **Effects of motor training on the recovery of manual dexterity after primary motor cortex lesion in macaque monkeys** *J Neurophysiol* **99**:773–786 [Google Scholar](#)
- Mussa-Ivaldi FA, Bizzi E (1995) **The Modular Organization of Motor Control: What Frogs Can Teach Us About Adaptive Learning** *IFAC Proceedings Volumes* **28**:413–418 [Google Scholar](#)
- Nath T, Mathis A, Chen AC, Patel A, Bethge M, Mathis MW (2019) **Using DeepLabCut for 3D markerless pose estimation across species and behaviors** *Nature Protocols* **14**:2152–2176 [Google Scholar](#)
- Nordez A, Gallot T, Catheline S, Guével A, Cornu C, Hug F (2009) **Electromechanical delay revisited using very high frame rate ultrasound** *J Appl Physiol* **106**:1970–1975 [Google Scholar](#)
- Nudo RJ, Wise BM, SiFuentes F, Milliken GW (1996) **Neural substrates for the effects of rehabilitative training on motor recovery after ischemic infarct** *Science* **272**:1791–1794 [Google Scholar](#)
- Overduin SA, d’Avella A, Roh J, Bizzi E (2008) **Modulation of Muscle Synergy Recruitment in Primate Grasping** *The Journal of Neuroscience* **28**:880–892 [Google Scholar](#)
- Petitot P, O’Reilly JX, O’Shea J (2018) **Towards a neuro-computational account of prism adaptation** *Neuropsychologia* **115**:188–203 [Google Scholar](#)
- Popa LS, Streng ML, Hewitt AL, Ebner TJ (2016) **The Errors of Our Ways: Understanding Error Representations in Cerebellar-Dependent Motor Learning** *Cerebellum* **15**:93–103 [Google Scholar](#)
- Power JD, Schlaggar BL (2017) **Neural plasticity across the lifespan** *Wiley Interdiscip Rev Dev Biol* **6** [Google Scholar](#)
- Ranganathan R, Adewuyi A, Mussa-Ivaldi FA (2013) **Learning to be lazy: exploiting redundancy in a novel task to minimize movement-related effort** *J Neurosci* **33**:2754–2760 [Google Scholar](#)
- Sattin D, Parma C, Lunetta C, Zulueta A, Lanzzone J, Giani L, Vassallo M, Picozzi M, Parati EA (2023) **An Overview of the Body Schema and Body Image: Theoretical Models, Methodological Settings and Pitfalls for Rehabilitation of Persons with Neurological Disorders** *Brain Sci* **13** [Google Scholar](#)
- Schärli A, Hecht H, Mast FW, Hossner EJ (2024) **How spotting technique affects dizziness and postural stability after full-body rotations in dancers** *Hum Mov Sci* **95**:103211 [Google Scholar](#)

- Schultz TW, Swett JE (1972) **The fine structure of the Golgi tendon organ** *J Neurocytol* **1**:1–26 [Google Scholar](#)
- Slawinska U, Kasicki S (2002) **Altered Electromyographic Activity Pattern of Rat Soleus Muscle Transposed into the Bed of Antagonist Muscle** *The Journal of Neuroscience* **22**:5808–5812 [Google Scholar](#)
- Smith MA, Shadmehr R (2005) **Intact ability to learn internal models of arm dynamics in Huntington’s disease but not cerebellar degeneration** *J Neurophysiol* **93**:2809–2821 [Google Scholar](#)
- Smith MA, Ghazizadeh A, Shadmehr R (2006a) **Interacting Adaptive Processes with Different Timescales Underlie Short-Term Motor Learning** *PLOS Biology* **4**:e179 [Google Scholar](#)
- Smith MA, Ghazizadeh A, Shadmehr R (2006b) **Interacting adaptive processes with different timescales underlie short-term motor learning** *PLoS Biol* **4**:e179 [Google Scholar](#)
- Smith SA, Mitchell JH, Garry MG (2006c) **The mammalian exercise pressor reflex in health and disease** *Exp Physiol* **91**:89–102 [Google Scholar](#)
- Sperry RW (1940) **The functional results of muscle transposition in the hind limb of the rat** *Journal of Comparative Neurology* **73**:379–404 [Google Scholar](#)
- Sperry RW (1942) **Transplantation of motor nerves and muscles in the forelimb of the rat** *Journal of Comparative Neurology* **76** [Google Scholar](#)
- Sugita Y (1996) **Global plasticity in adult visual cortex following reversal of visual input** *Nature* **380**:523–526 [Google Scholar](#)
- Takei T, Seki K (2010) **Spinal interneurons facilitate coactivation of hand muscles during a precision grip task in monkeys** *J Neurosci* **30**:17041–17050 [Google Scholar](#)
- Takei T, Confais J, Tomatsu S, Oya T, Seki K (2017) **Neural basis for hand muscle synergies in the primate spinal cord** *Proceedings of the National Academy of Sciences* **114**:8643–8648 [Google Scholar](#)
- Takeuchi N, Izumi S (2012) **Maladaptive plasticity for motor recovery after stroke: mechanisms and approaches** *Neural Plast* **2012**:359728 [Google Scholar](#)
- Tannenbaum J, Bennett BT (2015) **Russell and Burch’s 3Rs then and now: the need for clarity in definition and purpose** *J Am Assoc Lab Anim Sci* **54**:120–132 [Google Scholar](#)
- Thompson AK, Wolpaw JR (2014) **Operant conditioning of spinal reflexes: from basic science to clinical therapy** *Front Integr Neurosci* **8**:25 [Google Scholar](#)
- Trevartha KM, Garcia A, Wolpert DM, Flanagan JR (2014) **Fast but fleeting: adaptive motor learning processes associated with aging and cognitive decline** *J Neurosci* **34**:13411–13421 [Google Scholar](#)
- Vallbo AB, Hagbarth KE, Torebjörk HE, Wallin BG (1979) **Somatosensory, proprioceptive, and sympathetic activity in human peripheral nerves** *Physiol Rev* **59**:919–957 [Google Scholar](#)
- van Vliet P, Carey LM, Turton A, Kwakkel G, Palazzi K, Oldmeadow C, Searles A, Lavis H, Middleton S, Galloway M, Dimech-Betancourt B, O’Keefe S, Tavener M (2023) **Task-specific**

training versus usual care to improve upper limb function after stroke: the “Task-AT Home” randomised controlled trial protocol *Frontiers in Neurology* **14** [Google Scholar](#)

Vanhoof MJM, van Leeuwen T, Galletta L, Vereecke EE (2021) **The forearm and hand musculature of semi-terrestrial rhesus macaques (*Macaca mulatta*) and arboreal gibbons (fam. Hylobatidae). Part II. Quantitative analysis** *Journal of Anatomy* **238**:321–337 [Google Scholar](#)

Villalta JI, Landi SM, Fló A, Della-Maggiore V (2013) **Extinction Interferes with the Retrieval of Visuomotor Memories Through a Mechanism Involving the Sensorimotor Cortex** *Cerebral Cortex* **25**:1535–1543 [Google Scholar](#)

Walker JG, Jackson HJ, Littlejohn GO (2004) **Models of adjustment to chronic illness: using the example of rheumatoid arthritis** *Clin Psychol Rev* **24**:461–488 [Google Scholar](#)

Wester K, Hove LM, Barndon R, Craven AR, Hugdahl K (2018) **Cortical Plasticity After Surgical Tendon Transfer in Tetraplegics** *Front Hum Neurosci* **12**:234 [Google Scholar](#)

Winterbottom L, Nilsen DM (2024) **Motor Learning Following Stroke: Mechanisms of Learning and Techniques to Augment Neuroplasticity** *Phys Med Rehabil Clin N Am* **35**:277–291 [Google Scholar](#)

Wolpert DM (1997) **Computational approaches to motor control** *Trends Cogn Sci* **1**:209–216 [Google Scholar](#)

Yan Y, Sobinov AR, Bensmaia SJ (2022) **Prehension kinematics in humans and macaques** *J Neurophysiol* **127**:1669–1678 [Google Scholar](#)

Yumiya H, Larsen KD, Asanuma H (1979) **Motor readjustment and input-output relationship of motor cortex following cross-connection of forearm muscles in cats** *Brain Research* **177**:566–570 [Google Scholar](#)

Zajac FE (1992) **How musculotendon architecture and joint geometry affect the capacity of muscles to move and exert force on objects: a review with application to arm and forearm tendon transfer design** *J Hand Surg Am* **17**:799–804 [Google Scholar](#)

Zeiler SR, Krakauer JW (2013) **The interaction between training and plasticity in the poststroke brain** *Curr Opin Neurol* **26**:609–616 [Google Scholar](#)

Zemková E (2022) **Physiological Mechanisms of Exercise and Its Effects on Postural Sway: Does Sport Make a Difference?** *Front Physiol* **13**:792875 [Google Scholar](#)

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Reviewer #1 (Public review):

Summary:

Many studies have investigated adaptation to altered sensorimotor mappings or to an altered mechanical environment. This paper asks a different but also important question in motor control and neurorehabilitation: how does the brain adapt to changes in the controlled plant? The authors addressed this question by performing a tendon transfer surgery in two monkeys during which the swapped tendons flexing and extending the digits. They then monitored changes in task performance, muscle activation and kinematics post-recovery over several months, to assess changes in putative neural strategies.

Strengths:

(1) The authors performed complicated tendon transfer experiments to address their question of how the nervous system adapts to changes in the organisation of the

neuromusculoskeletal system, and present very interesting data characterising neural (and in one monkey, also behavioural) changes post tendon transfer over several months.

(2) The fact that the authors had to employ to two slightly different tasks -one more artificial, the other more naturalistic- in the two monkeys and yet found qualitatively similar changes across them makes the findings more compelling.

(3) The paper is quite well written, and the analyses are sound, although some analyses could be improved (suggestions below).

Weaknesses:

(1) I think this is an important paper, paper but I'm puzzled about a tension in the results. On the one hand, it looks like the behavioural gains post-TT happen rather smoothly over time (Figure 5). On the other, muscle synergy activations changes abruptly at specific days (around day ~65 for Monkey A and around day ~45 for monkey B; e.g., Figure 6). How do the authors reconcile this tension? In other words, how do they think that this drastic behavioural transition can arise from what appears to be step-by-step, continuous changes in muscle coordination? Is it "just" subtle changes in movements/posture exploiting the mechanical coupling between wrist and finger movements combined with subtle changes in synergies and they just happen to all kick in at the same time? This feels to me the core of the paper and should be addressed more directly.

(2) The muscles synergy analyses, which are an important part of the paper, could be improved. In particular:

(2a) When measuring the cross-correlation between the activation of synergies, the authors should include error bars, and should also look at the lag between the signals.

(2b) Figure 7C and related figures, the authors state that the activation of muscle synergies revert to pre-TT patterns toward the end of the experiments. However, there are noticeable differences for both monkeys (at the end of the "task range" for synergy B for monkey A, and around 50 % task range for synergy B for monkey B). The authors should measure this, e.g., by quantifying the per-sample correlation between pre-TT and post-TT activation amplitudes. Same for Figures 8I,J, etc.

(2c) In Figures 9 and 10, the authors show the cross-correlation of the activation coefficients of different synergies; the authors should also look at the correlation between activation profiles because it provides additional information.

(2d) Figure 11: the authors talk about a key difference in how Synergy B (the extensor finger) evolved between monkeys post-TT. However, to me this figure feels more like a difference in quantity -the time course- than quality, since for both monkeys the aaEMG levels pretty much go back to close to baseline levels -even if there's a statistically significant difference only for Monkey B. What am I missing?

(2e) Lines 408-09 and above: The authors claim that "The development of a compensatory strategy, primarily involving the wrist flexor synergy (Synergy C), appears crucial for enabling the final phase of adaptation", which feels true intuitively and also based on the analysis in Figure 8, but Figure 11 suggests this is only true for Monkey A. How can these statements be reconciled?

(3) Experimental design: at least for the monkey who was trained on the "artificial task" (Monkey A), it would have been good if the authors had also tested him on naturalistic grasping, like the second monkey, to see to what extent the neural changes generalise across behaviours or are task-specific. Do the authors have some data that could be used to assessed this even if less systematically?

(4) Monkey's B behaviour pre-tendon transfer seems more variable than that of Monkey A (e.g., the larger error bars in Figure 5 compared to monkey A, the fluctuating cross-correlation between FDS pre and EDC post in Figure 6Q), this should be quantified to better ground the results since it also shows more variability post-TT.

(5) Minor: Figure 12 is interesting and supports the idea that monkeys may exploit the biomechanical coupling between wrist and fingers as part of their function recovery. It would be interesting to measure whether there is a change in such coupling (tenodesis) over time, e.g., by plotting change in wrist angle vs change in MCP angle as a scatter plot (one dot per trial), and in the same plot show all the days, colour coded by day. Would the relationship remain largely constant or fluctuate slightly early on? I feel this analysis could also help address my point (1) above.

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Reviewer #2 (Public review):

Summary:

This study tackles an important question for both basic science understanding and translational relevance - how does the nervous system learn to control a changing body? Of course, all bodies change slowly over time, including basic parameters like size and weight distribution, but many types of diseases and injuries also alter the body and require neural adaptation to sustain normal control. A dramatic example from the clinic is the use of tendon transfer surgery in patients with near tetraplegia that allows them to use more proximal arm muscles to control the hand. Here, the authors sought to ask what strategies may be used when an animal adapts its motor control in response to tendon transfer. They focus on whether recovered functions leverage fractionated control over each muscle separately or, alternatively, whether there is evidence for modular control in which pre-existing synergies are recruited differently after the surgery. Overall, this work is very promising and advances the use of tendon transfer in animal models as a powerful way to study motor control flexibility, but the incomplete data and difficulty comparing between the two subjects mean that evidence is lacking for some of the conclusions.

Strengths:

A major strength of this paper is its motivating idea of using tendon transfer between flexor and extensor muscles in non-human primate wrist control to ask what adaptations are possible, how they evolve over time, and what might be the underlying neural control strategies. This is a creative and ambitious approach. Moreover, these surgeries are likely very challenging to do properly, and the authors rigorously documented the effectiveness of the transfer, particularly for Monkey A.

The results are promising, and there are two very interesting findings suggested by the data. First, when a single muscle out of a related group is manipulated, there is aberrant muscle activity detected across related muscles that are coordinated with each other and impacted as a group. For example, when the main finger extensor muscle now becomes a flexor, the timing of its activation is changed, and this is accompanied by similar changes in a more minor finger extensor as well as in wrist extensor muscles. This finding was observed in both monkeys and likely reflects a modular adaptive response. Second, there is a biphasic response in the weeks following injury, with an early phase in which the magnitude of an extensor synergy was increased and the timing of flexor and extensor recruitment was altered, followed by a later phase in which the timing and overall magnitude are restored.

Weaknesses:

The most notable weakness of the study is the incompleteness of the data. Monkey A has excellent quality EMG in all relevant muscles, but no analysis of video data, while Monkey B has some video data kinematics and moderate quality EMG, but the signal in the transferred FDS muscle was lost. These issues could be overcome by aligning data between the two monkeys, but the behavior tasks performed by each monkey are different, and so are the resulting muscle synergies detected (e.g., for synergies C and D), and different timepoints were analyzed in each monkey. As a result, it is difficult to make general conclusions from the study, and it awaits further analysis or the addition of another subject.

A second weakness is the insufficient analysis of the movements themselves, particularly for Monkey A. The main metrics analyzed were the time from task engagement (touch) to action onset and the time spent in an off-target location - neither of these measures can be related directly to muscle activity or the movement. Since the authors have video data for both monkeys, it is surprising that it was not used to extract landmarks for kinematic analysis, or at least hand/endpoint trajectory, and how it is adjusted over time. Adding more behavior data and aligning it with the EMG data would be very helpful for characterizing motor recovery and is needed to support conclusions about underlying neural control strategies for functional improvement.

Considering specific conclusions, the statement that the monkeys learned to use "tenodesis" over time by increasing activation of a wrist flexor muscle synergy does not seem to be fully supported by the data. Monkey A data includes EMG for two wrist flexors and a clear wrist flexor synergy, but it seems that, when comparing baseline and the final post-surgery timepoints, the main change is decreased activity around grasp after tendon transfer (at 0% of the task range if I understand this correctly) (Figure 8E and Figure S2-H vs R and -I vs S). It is clear that Monkey B increases the flexion of the wrist joint over time from the kinematic data, but the activity pattern in the only recorded wrist flexor (PL) doesn't change much with time (Figure S2-AN) and this monkey does not have a clear wrist flexor synergy (PL is active in the flexor synergy A while synergy C mainly reflects deltoid activity). Given these issues, it is not clear how to align the EMG and kinematic data and interpret these findings.

A more minor point regarding conclusions: statements about poor task performance and high energy expenditure being the costs that drive exploration for a new strategy are speculative and should be presented as such. Although the monkeys did take longer to complete the tasks after the surgery, they were still able to perform it successfully and in less than a second and no measurements of energy expenditure were taken.

A small concern is whether the tendon transfer effect may fail over time, either due to scar tissue formation or tendon tearing, and it would be ideal if the integrity of the intervention were re-assessed at the end of the study.

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Reviewer #3 (Public review):

Summary:

In this study, Philipp et al. investigate how a monkey learns to compensate for a large, chronic biomechanical perturbation - a tendon transfer surgery, swapping the actions of two muscles that flex and extend the fingers. After performing the surgery and confirming that the muscle actions are swapped, the authors follow the monkeys' performance on grasping tasks over several months. There are several main findings:

(1) There is an initial stage of learning (around 60 days), where monkeys simply swap the activation timing of their flexors and extensors during the grasp task to compensate for the

two swapped muscles.

(2) This is (seemingly paradoxically) followed by a stage where muscle activation timing returns almost to what it was pre-surgery, suggesting that monkeys suddenly swap to a new strategy that is better than the simple swap.

(3) Muscle synergies seem remarkably stable through the entire learning course, indicating that monkeys do not fractionate their muscle control to swap the activations of only the two transferred muscles.

(4) Muscle synergy activation shows a similar learning course, where the flexion synergy and extension synergy activations are temporarily swapped in the first learning stage and then revert to pre-surgery timing in the second learning stage.

(5) The second phase of learning seems to arise from making new, compensatory movements (supported by other muscle synergies) that get around the problem of swapped tendons.

Strengths:

This study is quite remarkable in scope, studying two monkeys over a period of months after a difficult tendon-transfer surgery. As the authors point out, this kind of perturbation is an excellent testbed for the kind of long-term learning that one might observe in a patient after stroke or injury, and provides unique benefits over more temporary perturbations like visuomotor transformations and studying learning through development. Moreover, while the two-stage learning course makes sense, I found the details to be genuinely surprising--specifically the fact that: (1) muscle synergies continue to be stable for months after the surgery, despite now being maladaptive; and (2) muscle activation timing reverts to pre-surgery levels by the end of the learning course. These two facts together initially make it seem like the monkey simply ignores the new biomechanics by the end of the learning course, but the authors do well to explain that this is mainly because the monkeys develop a new kind of movement to circumvent the surgical manipulation.

I found these results fascinating, especially in comparison to some recent work in motor cortex, showing that a monkey may be able to break correlations between the activities of motor cortical neurons, but only after several sessions of coaching and training (Oby et al. PNAS 2019). Even then, it seemed like the monkey was not fully breaking correlations but rather pushing existing correlations harder to succeed at the virtual task (a brain-computer interface with perturbed control).

Weaknesses:

I found the analysis to be reasonably well considered and relatively thorough. However, I do have a few suggestions that I think may elevate the work, should the authors choose to pursue them.

First, I find myself wondering about the physical healing process from the tendon transfer surgery and how it might contribute to the learning. Specifically, how long does it take for the tendons to heal and bear forces? If this itself takes a few months, it would be nice to see some discussion of this.

Second, I see that there are some changes in the muscle loadings for each synergy over the days, though they are relatively small. The authors mention that the cosine distances are very small for the conserved synergies compared to distances across synergies, but it would be good to get a sense for how variable this measure is within synergy. For example, what is the cosine similarity for a conserved synergy across different pre-surgery days? This might help inform whether the changes post-surgery are within a normal variation or whether they reflect important changes in how the muscles are being used over time.

Last, and maybe most difficult (and possibly out of scope for this work): I would have ideally liked to see some theoretical modeling of the biomechanics so I could more easily understand what the tendon transfer did or how specific synergies affect hand kinematics before and after the surgery. Especially given that the synergies remained consistent, such an analysis could be highly instructive for a reader or to suggest future perturbations to further probe the effects of tendon transfer on long-term learning.

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Author response:

Thank you for the thorough assessment and insightful reviews of our manuscript, "Multi-timescale neural adaptation underlying long-term musculoskeletal reorganization." We are very encouraged by the positive evaluation – particularly the recognition of the study as "important" with "solid" evidence – and we appreciate the constructive feedback provided in the public reviews.

As requested, we would like to provide this provisional author response to accompany the first version of the Reviewed Preprint. While we plan to provide a detailed point-by-point response upon submission of the revised manuscript, this email outlines our overall revision plan based on the public reviews.

We found the reviewers' comments to be extremely helpful and largely aligned with our own assessment of areas for clarification and strengthening. We plan a full revision that will address all points raised.

Regarding Interpretations and Clarity:

Several comments focused on clarifying key interpretations. We agree with these suggestions and have already incorporated significant textual revisions into the manuscript to:

More explicitly articulate the proposed multi-timescale model that reconciles the smooth behavioral recovery with the abrupt neural shifts (addressing a core point from R1).

Refine the interpretation of the compensatory tenodesis strategy, clarifying the distinct neural implementations observed in each monkey and the crucial role of temporal re-timing versus amplitude scaling (addressing points from R1 and R2).

Correct our interpretation regarding the apparent differences in the "arms race" phenomenon, framing it more parsimoniously in terms of observational windows and individual adaptation rates (addressing R1).

Ensure consistent and unambiguous terminology (e.g., using "activation profiles") throughout the text and figure captions (addressing R1).

Carefully adjust language to distinguish between direct empirical findings and interpretations regarding concepts like energetic cost and the drivers of adaptation (addressing R2).

Explicitly address the potential confound of physical tendon healing, clarifying in the Methods and Discussion why our surgical technique allows us to interpret the findings primarily in terms of neural learning (addressing R3).

Regarding New Analyses and Data Presentation:

The reviewers also provided excellent suggestions for new analyses to enhance the rigor and depth of our findings. We plan to undertake these analyses for the full revision, including:

Adding measures of trial-to-trial variability (e.g., SEM envelopes) and time-lag analysis to our cross-correlation results (addressing R1).

Performing a point-by-point statistical comparison to better characterize the subtle differences between pre-surgery and final recovered synergy profiles (addressing R1).

Formally quantifying the baseline behavioral variability between the monkeys (addressing R1).

Creating a new kinematic plot visualizing the refinement of the tenodesis skill over time (addressing R1).

Establishing a baseline for normal day-to-day synergy variability by analyzing pre-surgery data (addressing R3).

Incorporating additional behavioral/kinematic data (pull times and grasp aperture) into Figure 5 to provide a clearer link between neural changes and functional recovery (addressing R2).

We have also noted the reviewers' suggestions regarding figure clarity and plan improvements where possible. We have already addressed some specific recommendations (e.g., elaborating captions for Figs 6 & 7, adding a supplementary table for muscle acronyms).

We plan to address the 'Recommendations for the authors' thoroughly during the preparation of the revised manuscript. We are very grateful for all these recommendations, as we are confident they will significantly improve the quality, clarity, and impact of our work. We hope that these comprehensive revisions might also strengthen the final eLife assessment.

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